STRUCTURAL STABILITY

THEORY AND IMPLEMENTATION

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CONTENTS

PREFACE

This book presents a simple, concise, and reasonably comprehensive introduction to the principles and theory of structural stability that are the basis for structural steel design and shows how they may be used in the solution of practical building frame design problems. It provides the necessary background for the transition for students of structural engineering from fundamental theories of structural stability of members and frames to practical design rules in AISC Specifications. It was written for upper level undergraduate or beginning graduate students in colleges and universities on the one hand, and those in engineering practice on the other.

The scope of the book is indicated by its contents. The concepts and principles of structural stability presented in Chapter 1 form the basis for the elastic and plastic theories of stability of members and frames which are discussed separately in Chapter 2 (Columns), Chapter 3 (Beam-Columns), Chapter 4 (Rigid Frames), and Chapter 5 (Beams). The energy and numerical methods of analyzing a structure for its stability limit load are described in Chapter 6.

Each of these later chapters sets out initially to state the basic principles of structural stability, followed by the derivation of the necessary basic governing differential equations based on idealized conditions. These classical solutions and their physical significance are then examined. The chapter goes on to show how these solutions are affected by the inelasticity of the material and imperfection of the structural member and system associated with a real structure, using both hand techniques and modern computer capabilities. It finally outlines some of the popularly used techniques by which this voluminous information may be utilized to provide design rules and calculation techniques suitable for design office use. In this way, the reader not only will obtain an understanding of the fundamental principles and theory of structural stability from an idealized elastic, perfect system, but also to an inelastic imperfect system that leads to the necessary links between the code rules, design office practice, and the actual structural system in the real world.

The continued rapid development in computer hardware and software in recent years has made it possible for engineers and designers to predict structural behavior quite accurately. The advancement in structural analysis techniques coupled with the increased understanding of structural behavior has made it possible for engineers to adopt the Limit States Design philosophy. A limit state is defined as a condition at which a structural system or its component ceases to perform its intended function under normal conditions (Serviceability Limit State) or failure under severe conditions (Ultimate Limit State). The recently published Load and Resistance Factor Design (LRFD) Specification by the American Institute of Steel Construction (AISC) is based on the limit states philosophy and thus represents a more rational approach to the design of steel structures.

This book is not therefore just another book that presents Timoshenko's basic elastic theory (S. P. Timoshenko and J. M. Gere, "Theory of Elastic Stability," McGraw-Hill, 1961), or Bleich's inelastic buckling theory (F. Bleich, "Buckling Strength of Metal Structures," McGraw-Hill, 1952), or Chen's numerical analysis (W. F. Chen and T. Atsuta, "Theory of Beam-Columns," two-volume, McGraw-Hill, 1976, 1977) in a new style. Instead it presents theory and principles of structural stability in its most up-to-date form. This volume includes not only the state-of-the-art methods in the analysis and design of columns as individual members and as members of a structure, but also an introduction to engineers as to how these new developments have been implemented as the stability design criteria for members and frames in AISC/LRFD Specification.

This book is based on a series of lectures that Professor Chen gave at Purdue University and Lehigh University under the general heading of "Structural Stability." The preparation of the 1985 T. R. Higgins Lectureship Award paper entitled "Columns with End Restraint and Bending in Load and Resistance Factor Design" for AISC Engineering Journal (3rd Quarter, Vol. 22, No. 3, 1985) inspired us to attempt to create a useful textbook for the undergraduate and beginning graduate students in structural engineering as well as practicing structural engineers who are less familiar with the stability design criteria of members and frames in the newly published LRFD Specification.

Professor Chen wishes to extend his thanks to AISC for the 1985 T. R. Higgins Lectureship Award, when the book began to take shape, to

Professor H. L. Michael of Purdue University for continuing support over many years, and to the graduate students, C. Cheng, L. Duan, and F. H. Wu, among others, for preparing the Answers to Some Selected Problems during their course work on Structural Stability in the spring semester of 1986 in the School of Civil Engineering at Purdue University.

December, 1986
West Lafayette, IN

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U_w = $\frac{1}{2}\int_0^L EC_w\left(\frac{d^2\gamma}{dz^2}\right)^2 dz$
= strain energy due to warping restraint torsion
V = $-W_{ext}$ = potential energy
W_{int} = $-U$ = work done by the internal resisting forces
W_{ext} = $-V$ = work done by the external applied forces
Π = $U + V$ = total potential energy

GEOMETRY AND DIMENSIONS

A = cross sectional area
b_f = flange width
C_w = warping constant
= $\frac{1}{2} I_f h^2$ for I section
d = depth
h = distance between centroid of flanges
I = Ar^2 = moment of inertia
I_f = moment of inertia of one flange
J = uniform torsional (or St. Venant) constant
= $\sum_{i=1}^{n} \frac{1}{3} b_i t_i^3$ for a thin-walled open section
L = length
r = $\sqrt{\frac{I}{A}}$ = radius of gyration
S = elastic section modulus
t = thickness
u = displacement in the X-direction
v = displacement in the Y-direction
W = $\frac{\pi}{L}\sqrt{\frac{EC_w}{GJ}}$
Z = plastic section modulus
ϕ = curvature
λ_b = $\sqrt{\frac{M_p}{M_{cr}}}$ = beam slenderness parameter
λ_c = $\sqrt{\frac{P_y}{P_{ek}}} = \frac{KL}{\pi r}\sqrt{\frac{P_y}{E}}$ = column slenderness parameter
γ = angle of twist

MATERIAL PARAMETERS

E = Young's modulus
= 29,000 ksi for steel

E_{eff} = effective modulus
E_r = reduced modulus
E_t = tangent modulus
F_y, σ_y = yield stress
G = shear modulus
= $\dfrac{E}{2(1+\nu)} = 11{,}200$ ksi for steel
ν = Poisson's ratio
= 0.3 for steel

STABILITY RELATED FACTORS

A_F = amplification factor
B_1 = $P-\delta$ moment amplification factor for beam-columns in LRFD
= $\dfrac{C_m}{1-\left(\dfrac{P}{P_{ek}}\right)} \geq 1.0$
B_2 = $P-\Delta$ moment amplification factor for beam-columns in LRFD
= $\dfrac{1}{1-\sum\left(\dfrac{P}{P_{ek}}\right)}$ or
= $\dfrac{1}{1-\sum\left(\dfrac{P\Delta_0}{HL}\right)}$
C_b = $\dfrac{M_{cr}}{M_{ocr}}$ = equivalent moment factor for beams
= $1.75+1.05\left(\dfrac{M_1}{M_2}\right)+0.3\left(\dfrac{M_1}{M_2}\right)^2 \leq 2.3$ in AISC Specifications for end moment case
= $\dfrac{12}{3\dfrac{M_1}{M_{max}}+4\dfrac{M_2}{M_{max}}+3\dfrac{M_3}{M_{max}}+2}$ for other loading conditions
(*see* Table 5.2b, p. 334)
C_m = equivalent moment factor for beam-columns
= $0.6-0.4\left(\dfrac{M_1}{M_2}\right) \geq 0.4$ in ASD for end moment case
= $0.6-0.4\left(\dfrac{M_1}{M_2}\right)$ in LRFD for end moment case

NOTATION

LOAD AND MOMENT

P = axial load

P_e = $\dfrac{\pi^2 EI}{L^2}$ = Euler buckling load

P_{cr} = elastic buckling load

P_{ek} = $\dfrac{\pi^2 EI}{(KL)^2}$

= elastic buckling load considering column end conditions

P_f = failure load by the elastic–plastic analysis

P_p = plastic collapse load or limit load by the simple plastic analysis

P_r = $P_e \dfrac{E_r}{E}$ = reduced modulus load

P_t = $P_e \dfrac{E_t}{E}$ = tangent modulus load

P_u = ultimate strength considering geometric imperfections and material plasticity

P_y = AF_y = yield load

M_a = amplified (design) moment

M_{cr} = elastic buckling moment

M_{ocr} = $\dfrac{\pi}{L}\sqrt{EI_y GJ}\sqrt{1+W^2}$, where $W^2 = \dfrac{\pi^2}{L^2}\left(\dfrac{EC_w}{GJ}\right)$

= elastic buckling moment under uniform moment

M_{eq} = $C_m M_2$ = equivalent moment

M_{ext} = moment at a section due to externally applied loads

M_{int} = internal resisting moment of the section

M_{m} = transition moment (in Plastic Design)

M_{n} = nominal flexural strength

M_{p} = ZF_y = plastic moment

M_{pcx} = $1.18M_{px}\left[1-\left(\frac{P}{P_y}\right)\right] \leq M_{px}$

for H-section about strong axis.

= plastic moment capacity about the strong axis considering the influence of axial load

M_{pcy} = $1.19M_{py}\left[1-\left(\frac{P}{P_y}\right)^2\right] \leq M_{py}$

for H-section about weak axis.

= plastic moment capacity about the weak axis considering the influence of axial load

M_{u} = ultimate moment capacity considering geometric imperfections and material plasticity

M_{y} = SF_y = yield moment

T_{sv} = $GJ\frac{d\gamma}{dz}$ = St. Venant (or uniform) torsion

T_{w} = $-EC_w\frac{d^3\gamma}{dz^3}$ = warping restraint (or non-uniform) torsion

σ = stress

σ_{ij} = stress tensor

ε = strain

ε_{ij} = strain tensor

ENERGY AND WORK

U = $\frac{1}{2}\int_v \sigma_{ij}\varepsilon_{ij}\,dv = U_a + U_b + U_{sv} + U_w = -W_{int}$

= strain energy of a linear elastic system

U_a = $\frac{1}{2}\int_0^L \frac{P^2}{EA}dz = \frac{1}{2}\int_0^L EA\left(\frac{du}{dz}\right)^2 dz$

= strain energy due to axial shortening

U_b = $\frac{1}{2}\int_0^L \frac{M^2}{EI}dz = \frac{1}{2}\int_0^L EI\left(\frac{d^2v}{dz^2}\right)^2 dz$

= strain energy due to bending

U_{sv} = $\frac{1}{2}\int_0^L \frac{T_{sv}^2}{GJ}dz = \frac{1}{2}\int_0^L GJ\left(\frac{d\gamma}{dz}\right)^2 dz$

= strain energy due to St. Venant torsion

	=	$1 + \psi \dfrac{P}{P_{ek}}$ = effective length factor
K	=	$\sqrt{\dfrac{P_e}{P_{ek}}}$ = effective length factor
r_i	=	load factors
ϕ	=	resistance factor
ϕ_b	=	resistance factor for flexure = 0.90
ϕ_c	=	resistance factor for compression = 0.85

Chapter 1

GENERAL PRINCIPLES

1.1 CONCEPTS OF STABILITY

When a change in the geometry of a structure or structural component under compression will result in the loss of its ability to resist loadings, this condition is called *instability*. Because instability can lead to a catastrophic failure of a structure, it must be taken into account when one designs a structure. To help engineers to do this, among other types of failure, a new generation of designing codes have been developed based on the concept of *limit states*.

In *limit states design*, the structure or structural component is designed against all pertinent limit states that may affect the safety or performance of the structure. Basically, there are two types of limit states: The first type, *Strength limit states*, deals with the performance of structures at their maximum load-carrying capacities. Examples of strength limit states include structural failure due to either the formation of a plastic collapse mechanism or to member or frame instability. *Serviceability limit states*, on the other hand, are concerned with the performance of structures under normal service conditions. Hence, they pertain to the appearance, durability, and maintainability of a structure. Examples of serviceability limit states include deflections, drift, vibration, and corrosion.

Stability, an important constituent of the strength limit states, is dealt with explicitly in the present American Institute for Steel Construction (AISC) limit state specification.[1] Although the importance of considering stability in design is recognized by most practicing engineers, the subject still remains perplexing to some. The reason for this perplexity is that the use of *first-order structural analysis*, which is familiar to most engineers, is not permissible in a stability analysis. In a true *stability analysis*, the change in geometry of the structure must be taken into account; as a consequence, equilibrium equations must be written based on the geometry of a structure that becomes deformed under load. This is known as the *second-order analysis*. The second-order analysis is further complicated by the fact that the resulting equilibrium equations are differential equations instead of the usual algebraic equations. Consequently, a mastery of differential calculus is a must before any attempt to solve these equations.

In what follows, we will explain the nature of structural stability and ways to analyze it accurately.

The concept of stability is best illustrated by the well-known example of a ball on a curved surface (Fig. 1.1). For a ball initially in equilibrium, a slight disturbing force applied to the ball on a concave surface (Fig. 1.1a) will displace the ball by a small amount, but the ball will return to its initial equilibrium position once it is no longer being disturbed. In this case, the ball is said to be in a *stable equilibrium*. If the disturbing force is applied to a ball on a convex surface (Fig. 1.1b) and then removed, the ball will displace continuously from, and never return to, its initial equilibrium position, even if the disturbance was infinitesimal. The ball in this case is said to be in an *unstable equilibrium*. If the disturbing force is

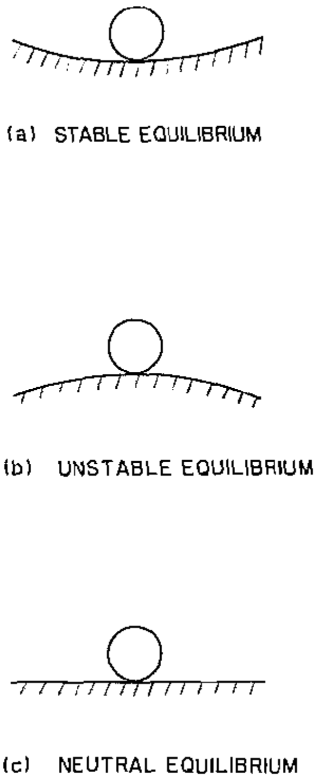


FIGURE 1.1 Stable, unstable, and neutral equilibrium

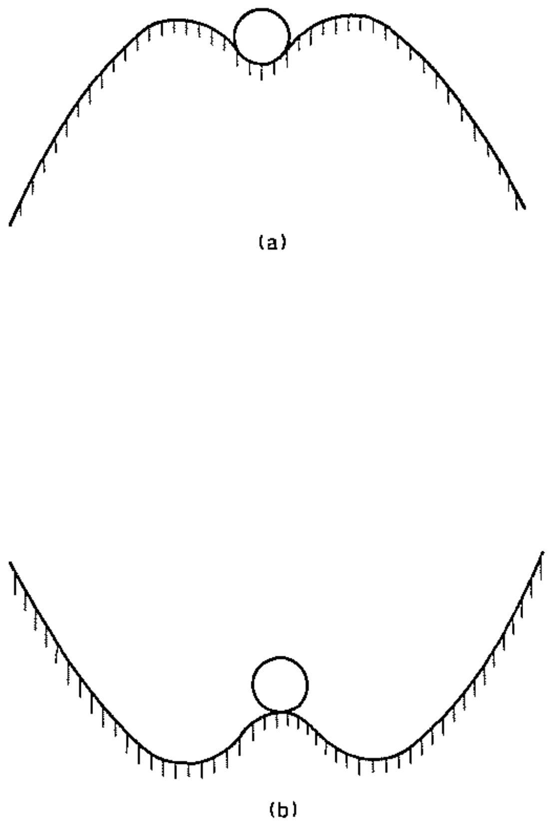


FIGURE 1.2 Effect of finite disturbance

applied to the ball on a flat surface (Fig. 1.1c), the ball will attain a new equilibrium position to which the disturbance has moved it and will stay there when the disturbance is removed. This ball is said to be in a *neutral equilibrium.*

Note that the definitions of stable and unstable equilibrium in the preceding paragraph apply only to cases in which the disturbing force is very small. These will be our *definitions of stability.* However, keep in mind that it is possible for a ball, under certain conditions (Fig. 1.2), to go from one equilibrium position to another; for example, a ball that is "stable" under a small disturbance may go to an unstable equilibrium under a large disturbance (Fig. 1.2a), or vice versa (Fig. 1.2b).

The concept of stability can also be explained by considering a system's stiffness. For an n-degrees-of-freedom system, the forces and displacements of the system are related by a stiffness matrix or function. If this stiffness matrix or function is *positive definite,* the system is said to be stable. The transition of the system from a state of stable to neutral equilibrium or from a state of stable to unstable equilibrium is marked by the *stability limit point.* The tangent stiffness of the system vanishes at just this point. We shall use the principle of vanishing tangent stiffness to calculate the buckling load of a system in subsequent sections and chapters.

Stability of an elastic system can also be interpreted by means of the concept of *minimum total potential energy.* In nature, an elastic system always tends to go to a state in which the total potential energy is at a minimum. The system is in a stable equilibrium if any deviation from its initial equilibrium state will result in an increase in the total potential energy of the system. The system is in an unstable equilibrium if any deviation from its initial equilibrium state will result in a decrease in total potential energy. Finally, the system is in a neutral equilibrium if any deviation from its initial equilibrium state will produce neither an increase nor a decrease in its total potential energy. Because of this principle, the energy concept can be used to find the buckling load of an elastic system. The elastic buckling analysis by energy method will be discussed in Chapter 6.

1.2 TYPES OF STABILITY

Stability of structures under compressive forces can be grouped into two categories: (1) instability that associates with a *bifurcation of equilibrium* (Fig. 1.3a); and (2) instability that associates with a *limit or maximum load* (Fig. 1.3b).

1.2.1 Bifurcation Instability

This type of instability is characterized by the fact that as the compressive load increases, the member or system that originally deflects in the direction of the applied loads suddenly deflects in a different direction. The point of transition from the usual deflection mode under loads to an alternative deflection mode is referred to as the *point of bifurcation of equilibrium.* The load at the point of bifurcation of equilibrium is called the *critical load.* The deflection path that exists before bifurcation is known as the *primary* or *fundamental path* and the deflection path that exists subsequent to bifurcation is known as the *secondary* or *postbuckling path* (Fig. 1.3a). Examples of this type of instability include the buckling of geometrically perfect columns loaded axially, buckling of thin plates subjected to in-plane compressive forces and buckling of rings subjected to radial compressive forces.

Depending on the nature of the postbuckling paths, two types of bifurcation can be identified: symmetric bifurcation and asymmetric bifurcation (Fig. 1.4).

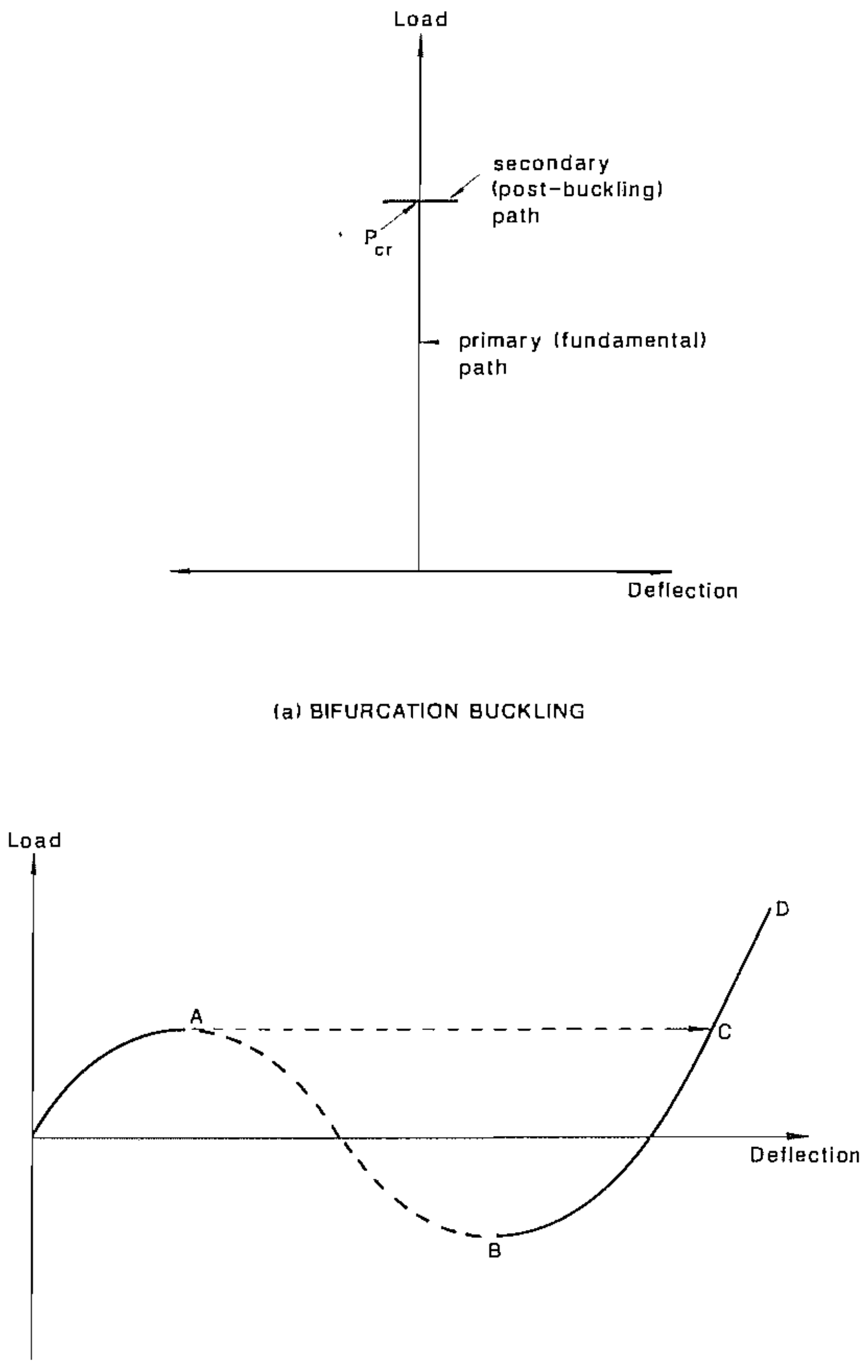


FIGURE 1.3 Bifurcation and limit point buckling

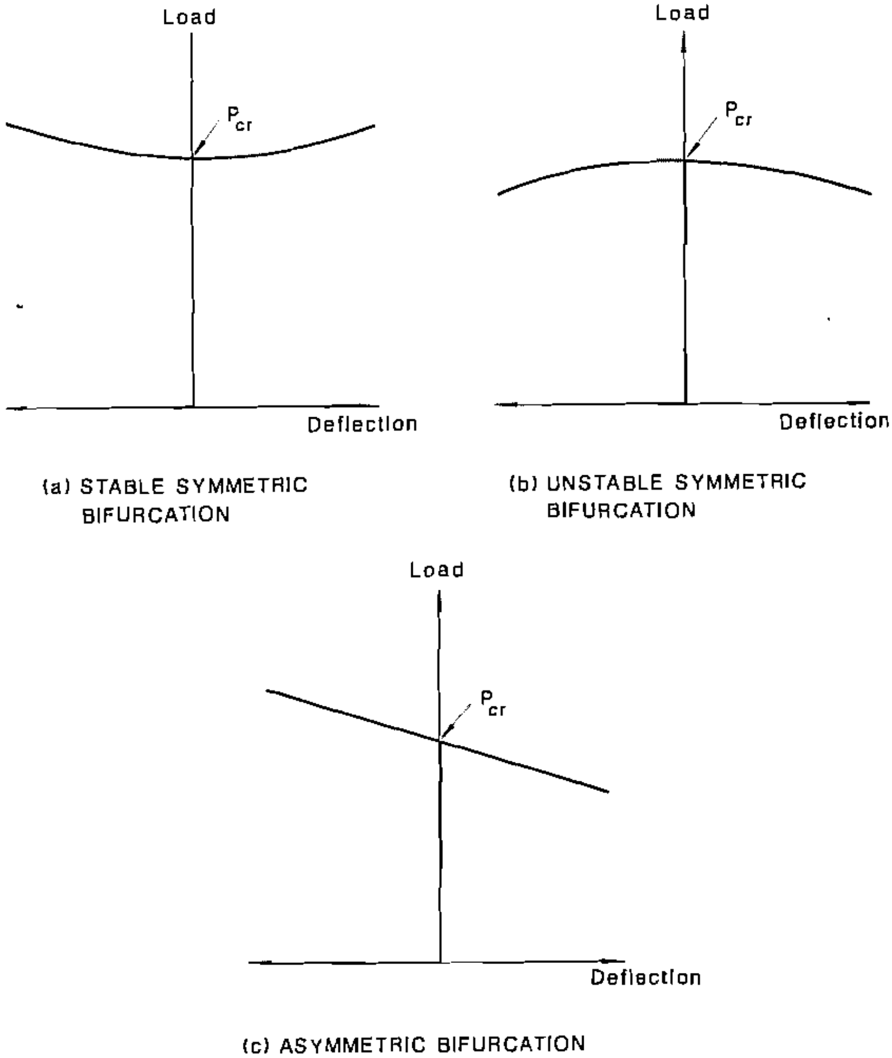


FIGURE 1.4 Postbuckling behavior

Symmetric Bifurcation

For symmetric bifurcation, the postbuckling paths are symmetric about the load axis. If the postbuckling paths rise above the critical load, the system is said to exhibit a *stable symmetric bifurcation*. For such a system, the load that is required to maintain equilibrium subsequent to buckling increases with increasing deformation, as shown in Fig. 1.4a. Examples of structures exhibiting stable postbuckling behavior include axially loaded elastic columns and in-plane loading of the thin elastic plates (Fig. 1.5). If the postbuckling paths drop below the critical load, the system is said to

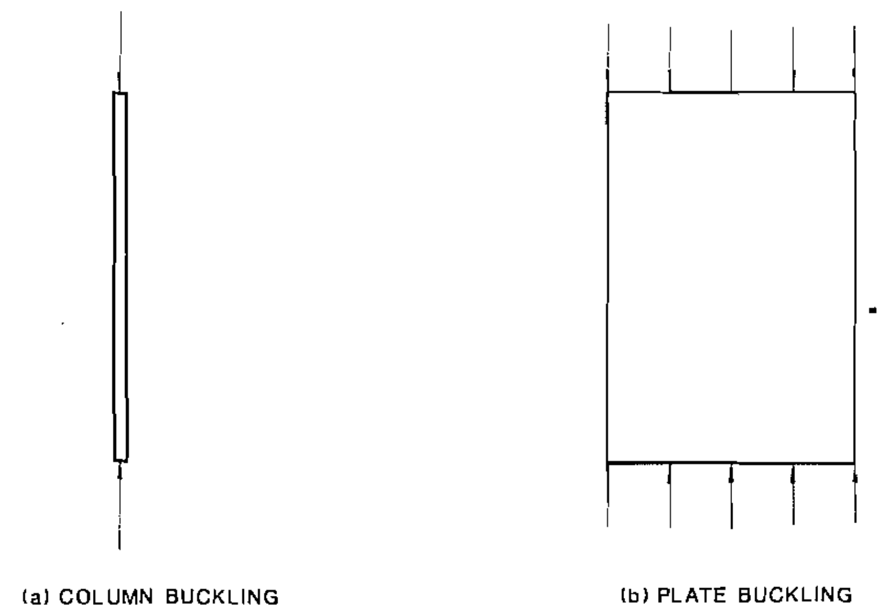


FIGURE 1.5 Examples of stable symmetric buckling

exhibit an *unstable symmetric bifurcation.* For such a system, the load that is required to maintain equilibrium subsequent to buckling decreases with increasing deflection as shown in Fig. 1.4b. The guyed tower shown in Fig. 1.6 is an example of a structure that exhibits an unstable postbuckling behavior. As the tower buckles and deflects, some of the cables are stretched, resulting in a detrimental pulling force on the tower.

Asymmetric Bifurcation

Figure 1.4c shows schematically the asymmetric bifurcation behavior of a system. For such a system, the load that is required to maintain equilibrium subsequent to buckling may increase or decrease with increasing deflection depending on the direction in which the structure deflects after buckling. The simple frame shown in Fig. 1.7 is an example of a structure that exhibits an *asymmetric postbuckling behavior.* If the frame buckles according to Mode 1, the shear force V induced in the beam will counteract the applied force P in the column. As a result, the load required to maintain equilibrium after buckling will increase with increasing deflection. On the other hand, if the frame buckles according to Mode 2, the shear force V induced in the beam will intensify the applied force P in the column. As a result, the load required to maintain equilibrium after buckling will decrease with increasing deflection.

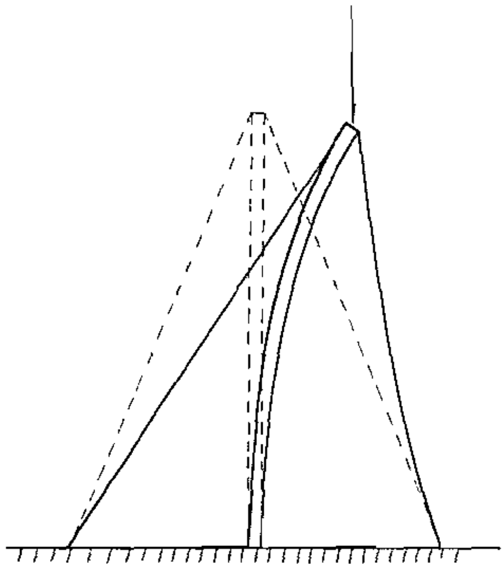


FIGURE 1.6 Example of unstable symmetric buckling

1.2.2 Limit-Load Instability

This type of instability is characterized by the fact that there is only a single mode of deflection from the start of loading to the limit or maximum load (Fig. 1.3b). Examples of this type of instability are buckling of shallow arches and spherical caps subjected to uniform external pressure (Fig. 1.8). For this type of buckling, once the limit load is reached (Point A on the curve of Fig. 1.3b), the system will "*snap through*" from Point A to Point C, because the equilibrium path AB is an unstable one. This unstable equilibrium path will never be encountered under a load controlled testing condition, but it does exist and can be observed under a displacement controlled testing condition. The phenomenon in which a visible and sudden jump from one equilibrium configuration to another nonadjacent equilibrium configuration upon reaching the limit load is referred to as *snap-through buckling*.

Another type of buckling that is unique to shells under compressive forces (Fig. 1.9a) is referred to by Libove in reference 2 as *finite-disturbance buckling*. For this type of buckling the compressive force required to maintain equilibrium drops considerably as the structure buckles after reaching the critical load as shown in curve (i) of Fig. 1.9b. In fact, in reality the theoretical critical load will never be reached because of imperfections. The slightest imperfections in such structures will reduce the critical load tremendously and so curve (ii) in the figure will be more representative of the actual buckling behavior of the real structure.

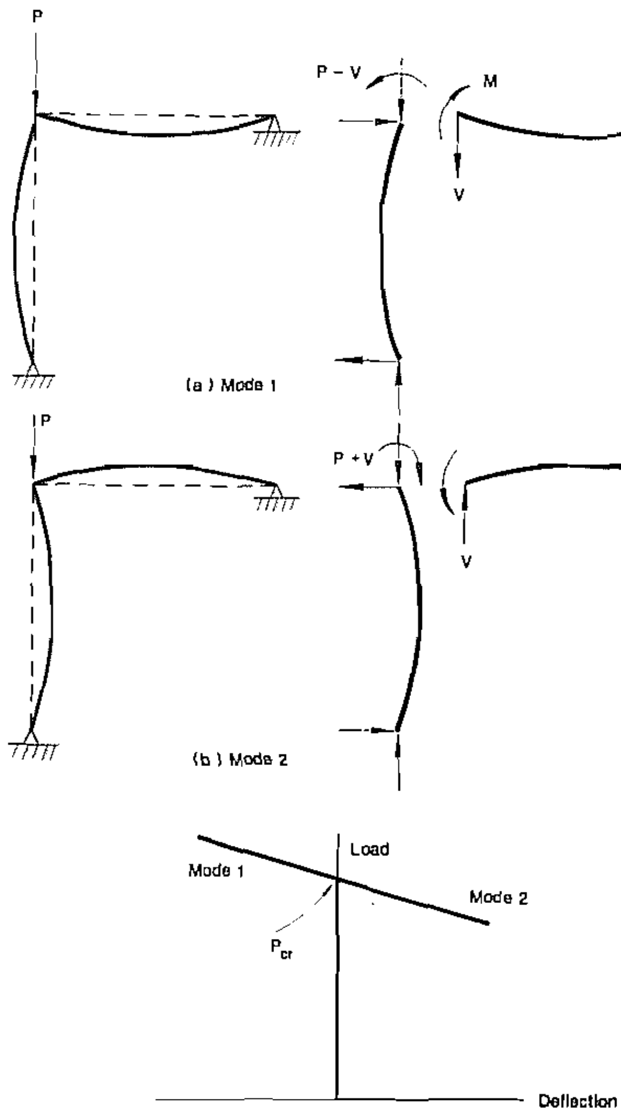


FIGURE 1.7 Example of asymmetric buckling

Finite disturbance buckling has the features of both bifurcation buckling and snap-through buckling. It resembles the former in that the shell deflects in one mode before the critical load is reached, but then deflects in a distinctly different mode after the critical load is reached. It resembles the latter in that a slight disturbance may trigger a jump from the original equilibrium configuration that exists before the critical load to a nonadjacent equilibrium configuration at finite deflections as indicated by the dotted line in the figure.

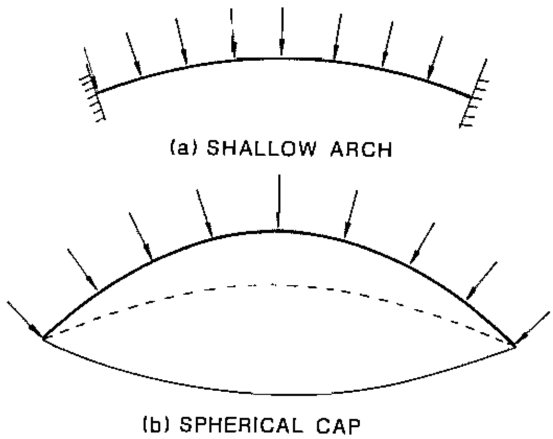
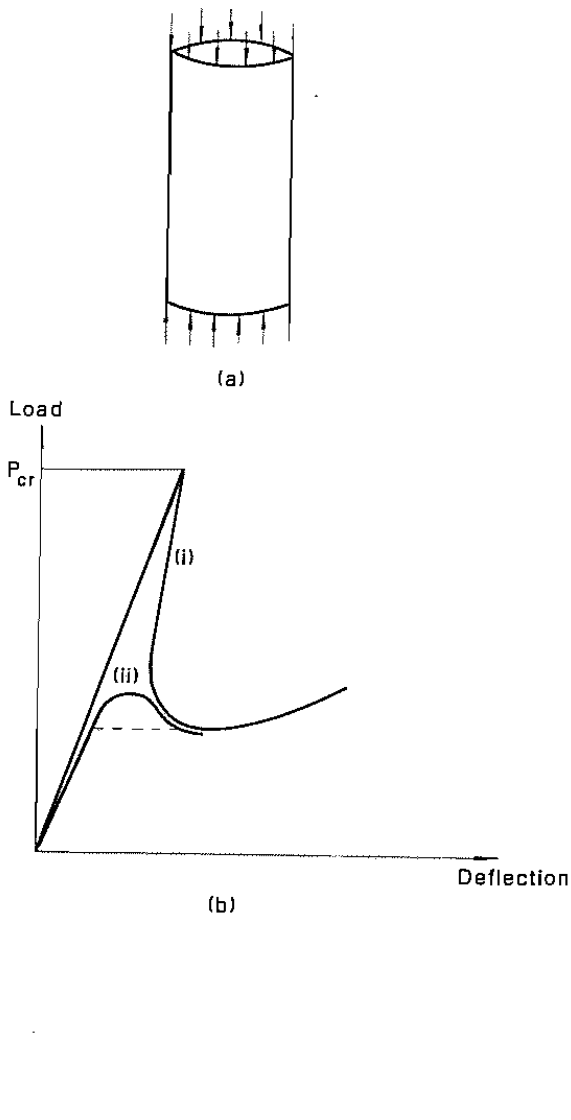


FIGURE 1.8 Examples of limit point buckling

FIGURE 1.9 Shell buckling



1.3 METHODS OF ANALYSES IN STABILITY

The concept of stability as described in Section 1.1 can be used to determine the critical conditions of an elastic system that is susceptible to instability.

Bifurcation Approach

The first approach is called the *bifurcation approach*. In this approach, the state at which two or more different but adjacent equilibrium configurations can exist is sought by an *eigenvalue analysis*. The lowest load that corresponds to this state is the critical load of the system. At the critical load, equilibrium can be maintained with alternative deflection modes that are infinitesimally close to one another.

To determine the critical load using the bifurcation approach, it is necessary to identify all possible equilibrium configurations the system can assume at the bifurcation load. This can best be accomplished by specifying a set of generalized displacements to describe all the possible displaced configurations of the system. If n parameters are required to describe the various modes of deflections, the system is said to have n degrees of freedom. For an n-degrees-of-freedom system, the determinant of the $n \times n$-system-stiffness matrix, which relates the generalized forces to the generalized displacements of the system, is a measure of the stiffness of the system. At the critical load, the tangent stiffness of the system vanishes. Thus, by setting the determinant of the system's-tangent-stiffness matrix equal to zero, the system's critical conditions can be identified.

The bifurcation approach is also known as the *eigenvalue approach,* because the technique used is identical to that used in the linear algebra for finding eigenvalues of a matrix. The critical conditions are represented by the eigenvalues of the system's stiffness matrix and the displaced configurations are represented by the eigenvectors. The lowest eigenvalue is the critical load of the system. The bifurcation or eigenvalue approach is an idealized mathematical approach to determine the critical conditions of a *geometrically perfect* system. If geometrical imperfections are present, deflection will commence at the beginning of loading. The problem then becomes a *load-deflection* rather than a bifurcation problem. For a load-deflection problem, the bifurcation approach cannot be applied.

Energy Approach

Another way to determine the critical conditions of a system is the *energy approach*. For an *elastic* system subjected to *conservative forces,* the total potential energy of the system can be expressed as a function of a set of generalized displacements and the external applied forces. The term "conservative forces" used here are those forces whose potential energy is dependent only on the final values of deflection, not the specific paths to reach these final values. If the system is in equilibrium, its total potential energy must be stationary. Thus, by setting the first derivative of the total potential energy function with respect to each generalized displacement equal to zero, we can identify the equilibrium conditions of the system. The critical load can then be calculated from the equilibrium equations.

Please note that by setting the first derivative of the total potential energy function equal to zero, we can only identify the equilibrium conditions of the system. To determine whether the equilibrium is stable or unstable, we must investigate higher order derivatives of the total potential energy function.

Dynamic Approach

The critical load of an elastic system can also be obtained by the *dynamic approach*. Here, a system of equations of motion governing the small free vibration of the system is written as a function of the generalized displacements and the external applied force. The critical load is obtained as the level of external force when the motion ceases to be bounded. The equilibrium is stable if a slight disturbance causes only a slight deviation of the system from its original equilibrium position and if the magnitude of the deviation decreases when the magnitude of the disturbance decreases. The equilibrium is unstable if the magnitude of motion increases without bound when subjected to a slight disturbance. The use of the dynamic approach requires a prerequisite of structural dynamics. This is beyond the scope of this book. However, the use of the other two approaches to determine the critical loads will be illustrated in the following sections. In the following examples, both the small and large deflection analyses will be used to demonstrate the significance and physical implications of each analysis.

1.4 ILLUSTRATIVE EXAMPLES—SMALL DEFLECTION ANALYSIS

In this section, the stability behavior of some simple structural models will be investigated in the context of a small deflection analysis by using both the bifurcation and energy approaches.

1.4.1 Rigid Bar Supported by a Rotational Spring

Consider the simple spring–bar system shown in Fig. 1.10a. The bar is assumed to be rigid, and the only possible mode of displacement for the system is the rigid body rotation of the bar about the pinned end as shown in Fig. 1.10b. The pinned end is supported by a linear rotational

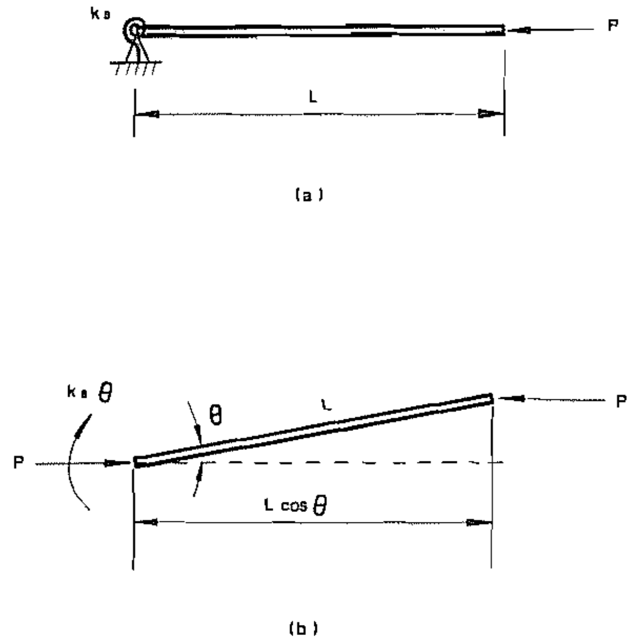


FIGURE 1.10 Rotational spring-supported rigid bar system (small deflection analysis)

spring of stiffness k_s. When the rigid bar is perfectly horizontal, the spring is in an unstrained state, and we shall denote any rotational displacement of the bar from this horizontal position by the angle θ.

The system will become unstable and buckle when P reaches its critical value, P_{cr}. We shall use the two methods already discussed to determine this initial value.

Bifurcation Approach

Assuming the rotational displacement θ is small, then the equilibrium condition of the bar at its displaced configuration can be written in the simple form as

$$k_s\theta - PL\theta = 0 \tag{1.4.1}$$

This equation is always satisfied for $\theta = 0$. The horizontal position or $\theta = 0$ is therefore a trivial solution. For a nontrivial solution, we must have

$$P = P_{cr} = \frac{k_s}{L} \tag{1.4.2}$$

Equation (1.4.2) indicates that when the applied force P reaches the value of k_s/L, the system will buckle. At this critical load, equilibrium for the bar is possible in both the original horizontal and slightly deflected positions.

Energy Approach

In using the energy approach to determine P_{cr}, one must write an expression for the total potential energy comprising the strain energy and the potential energy of the system. The strain energy stored in the system as the bar assumes its slightly deflected configuration is equal to the strain energy of the spring:

$$U = \tfrac{1}{2}k_s\theta^2 \tag{1.4.3}$$

The potential energy of the system is the potential energy of the external force and it is equal to the negative of the work done by the external force on the system:

$$V = -PL(1 - \cos\theta) \tag{1.4.4}$$

The term $L(1 - \cos\theta)$ represents the horizontal distance traveled by P as the bar rotates.

The total potential energy is the sum of the strain energy and the potential energy of the system:

$$\Pi = U + V = \tfrac{1}{2}k_s\theta^2 - PL(1 - \cos\theta) \tag{1.4.5}$$

For equilibrium, the total potential energy must assume a stationary value. Thus, we must have

$$\frac{d\Pi}{d\theta} = 0$$

or

$$k_s\theta - PL\sin\theta = 0 \tag{1.4.6}$$

For small θ, $\sin\theta \approx \theta$, therefore, we have

$$k_s\theta - PL\theta = 0 \tag{1.4.7}$$

Equation (1.4.7) is the equilibrium equation of the bar-spring system. The same equilibrium equation [Eq. (1.4.1)] has been obtained by considering equilibrium of a free body of the bar. The nontrivial solution for Eq. (1.4.7) is thus the critical load of the system.

$$P = P_{cr} = \frac{k_s}{L} \tag{1.4.8}$$

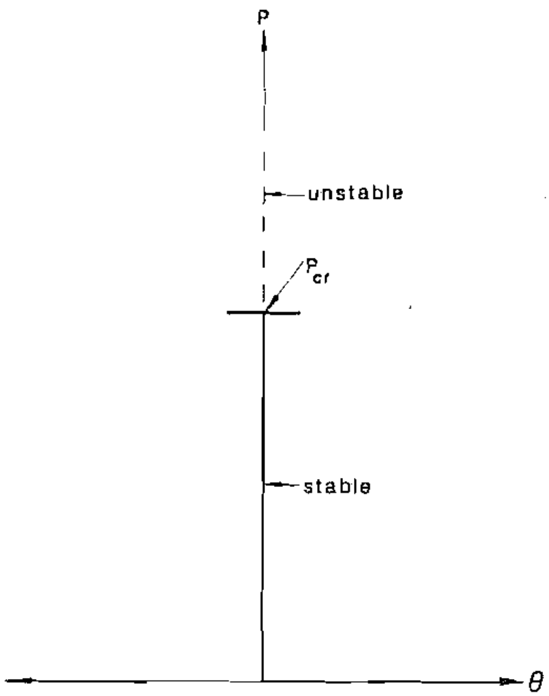
This value of the critical load is the same as that determined earlier by the bifurcation approach. Note that in the energy approach we can determine not only the critical load, P_{cr}, but also the nature of the equilibrium of this system. That is, we can further determine whether the equilibrium of the system is stable or unstable at various stages of loading.

To determine whether the equilibrium is stable or unstable for the system in its original ($\theta = 0$) position, we need to investigate the positiveness or negativeness of higher derivatives of the total potential energy function. For this problem, if we take the second derivative of the total potential energy function given in Eq. (1.4.5), we have

$$\frac{d^2\Pi}{d\theta^2} = k_s - PL \cos\theta \approx k_s - PL \tag{1.4.9}$$

Thus, for $P < P_{cr}$, $d^2\Pi/d\theta^2$ is positive. This indicates that the equilibrium is *stable*. For $P > P_{cr}$, $d^2\Pi/d\theta^2$ is negative, and the equilibrium condition is therefore *unstable*. This behavior is illustrated in Fig. 1.11, in which the applied force P is plotted as a function of θ for *small* values of θ. The solid line represents a stable equilibrium loading path and the dotted line represents an unstable equilibrium path. The bar is in a stable equilibrium in the horizontal ($\theta = 0$) position when $P < P_{cr}$, but becomes unstable in that position when $P > P_{cr}$. For $P = P_{cr}$, $d^2\Pi/d\theta^2 = 0$ in a

FIGURE 1.11 Stability behavior of spring-bar system (small deflection analysis)



small θ analysis and so no conclusion can be drawn. However, as will be demonstrated in the next section, the stability condition at the critical load ($P = P_{cr}$) can be investigated using the energy approach with a large θ analysis.

1.4.2 Rigid Bar Supported by a Translational Spring

Figure 1.12a shows a rigid bar hinged at one end and supported by a linear translational spring of stiffness k_s at the free end. The spring is assumed to be able to move freely in the horizontal direction but retains its vertical orientation as the bar deflects. The bar is subjected to a concentrated force P at the free end. Assuming the system is geometrically perfect, we shall determine the critical load of the system.

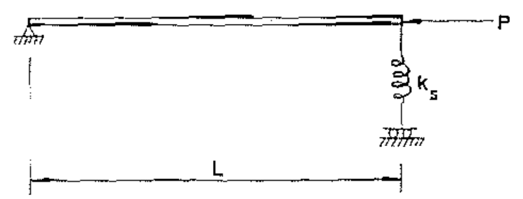
Bifurcation Approach

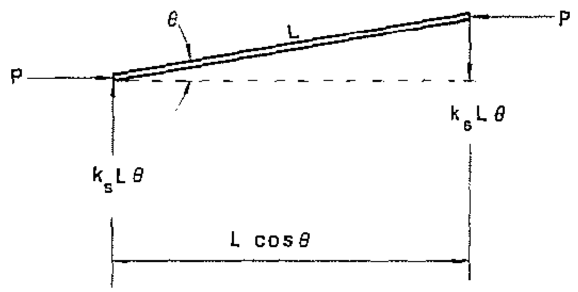
In Fig. 1.12b we see a slightly deflected position of the bar. Consideration of equilibrium of the bar gives

$$k_s L^2\theta - PL\theta = 0 \tag{1.4.10}$$

This condition is always satisfied by the trivial solution $\theta = 0$. For a

FIGURE 1.12 Translational spring-supported rigid bar system (small deflection analysis)





nontrivial solution, we must have

$$P = P_{cr} = k_s L \tag{1.4.11}$$

Energy Approach

The strain energy of the system at its deflected state is equal to the strain energy stored in the structural spring.

$$U = \tfrac{1}{2} k_s (L\theta)^2 \tag{1.4.12}$$

The potential energy of the system is equal to the negative of the work done by the applied force.

$$V = -PL(1 - \cos\theta) \tag{1.4.13}$$

The total potential energy is

$$\Pi = U + V = \tfrac{1}{2} k_s (L\theta)^2 - PL(1 - \cos\theta) \tag{1.4.14}$$

For equilibrium, the first derivative of Π with respect to θ must vanish.

$$\frac{d\Pi}{d\theta} = k_s L^2 \theta - PL \sin\theta = 0 \tag{1.4.15}$$

For small θ, Eq. (1.4.15) can be written as

$$k_s L^2 \theta - PL\theta = 0 \tag{1.4.16}$$

This equilibrium equation is identical to that of Eq. (1.4.10). Therefore, the nontrivial solution is

$$P = P_{cr} = k_s L \tag{1.4.17}$$

To investigate the nature of equilibrium of the system in its original ($\theta = 0$) position, we need to perform higher order derivatives of the total potential energy function. By taking the second derivative of the total potential energy function given in Eq. (1.4.14), we have

$$\frac{d^2\Pi}{d\theta^2} = k_s L^2 - PL\cos\theta \approx k_s L^2 - PL \tag{1.4.18}$$

Thus, for $P < P_{cr}$, $d^2\Pi/d\theta^2$ is positive, so the system is stable, but for $P > P_{cr}$, $d^2\Pi/d\theta^2$ is negative, so the system is unstable in its original ($\theta = 0$) position.

1.4.3 Two-Bar System

Consider the two-bar system shown in Fig. 1.13a. The two bars are linked together by a frictionless pin at C and the entire system is supported at three locations. The supports at B and D are hinged. The support at C is a spring with spring stiffness k_s. The bars are subjected to

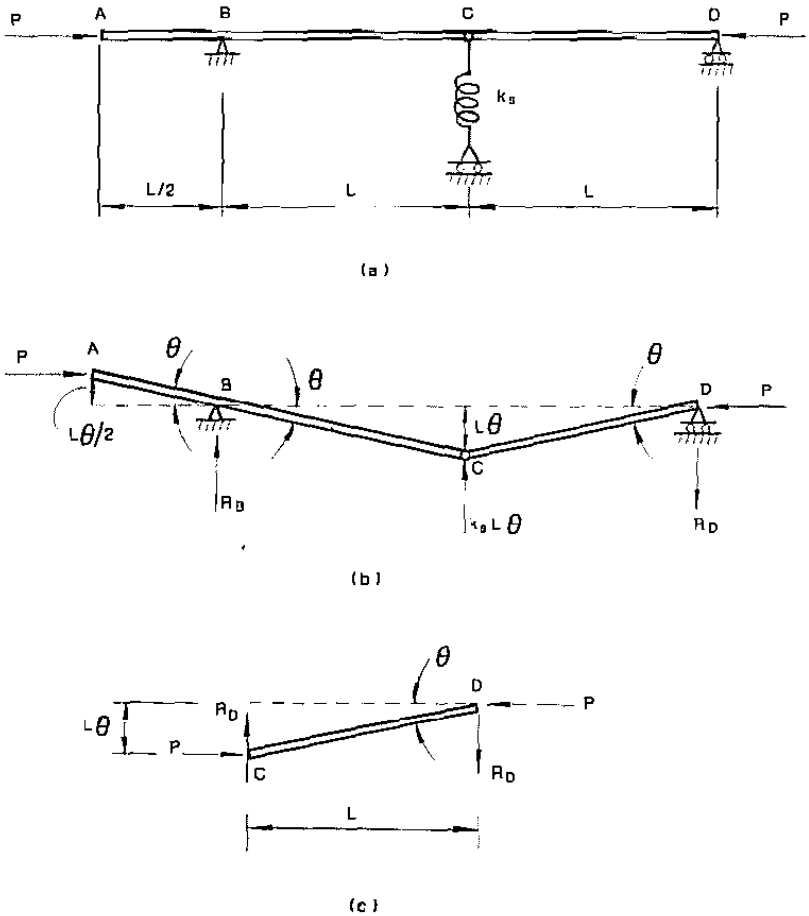


FIGURE 1.13 Two-bar system

a compressive force P at the ends. As P increases, a condition will be reached at which the bars will assume a slightly deflected position. This deflected position can be defined uniquely by a single parameter θ as shown in Fig. 1.13b.

Bifurcation Approach

Summing moments about B for the free body shown in Fig. 1.13b gives

$$P\left(\frac{L\theta}{2}\right) - k_s L\theta(L) + R_D(2L) = 0$$

from which, we solve for

$$R_D = \frac{2k_s L\theta - P\theta}{4} \tag{1.4.19}$$

Summing moment about C for the free body in Fig. 1.13c gives

$$R_D L - PL\theta = 0 \tag{1.4.20}$$

Upon substitution of Eq. (1.4.19) into Eq. (1.4.20), we obtain

$$\tfrac{1}{2}k_s L\theta - \tfrac{5}{4}P\theta = 0 \tag{1.4.21}$$

The nontrivial solution for the equilibrium equation (1.4.21) gives the critical load of the system as

$$P = P_{cr} = \tfrac{2}{5}k_s L \tag{1.4.22}$$

Energy Approach

The strain energy of the system is

$$U = \tfrac{1}{2}k_s(L\theta)^2 \tag{1.4.23}$$

The potential energy of the system is

$$\begin{aligned} V &= -P[L(1-\cos\theta) + L(1-\cos\theta) \\ &\quad + \tfrac{1}{2}L(1-\cos\theta)] \\ &= -P[\tfrac{5}{2}L(1-\cos\theta)] \end{aligned} \tag{1.4.24}$$

The total potential energy of the system is

$$\Pi = U + V = \tfrac{1}{2}k_s(L\theta)^2 - P[\tfrac{5}{2}L(1-\cos\theta)] \tag{1.4.25}$$

For equilibrium, we must have

$$\frac{d\Pi}{d\theta} = k_s L^2\theta - \tfrac{5}{2}PL\sin\theta = 0 \tag{1.4.26}$$

and for small θ, we have

$$k_s L^2\theta - \tfrac{5}{2}PL\theta = k_s L\theta - \tfrac{5}{2}P\theta = 0 \tag{1.4.27}$$

The critical load is obtained as the nontrivial solution of Eq. (1.4.27).

$$P = P_{cr} = \tfrac{2}{5}k_s L \tag{1.4.28}$$

The nature of equilibrium of the system in its original ($\theta = 0$) position can be studied by observing higher derivatives of the total potential energy function. By taking the second derivative of the total potential energy function, we have

$$\frac{d^2\Pi}{d\theta^2} = k_s L^2 - \tfrac{5}{2}PL\cos\theta \approx k_s L^2 - \tfrac{5}{2}PL \tag{1.4.29}$$

For $P < P_{cr}$, $d^2\Pi/d\theta^2$ is positive, so the system is stable. For $P > P_{cr}$, $d^2\Pi/d\theta^2$ is negative, so the system is unstable in its original undeflected position.

1.4.4 Three-Bar System

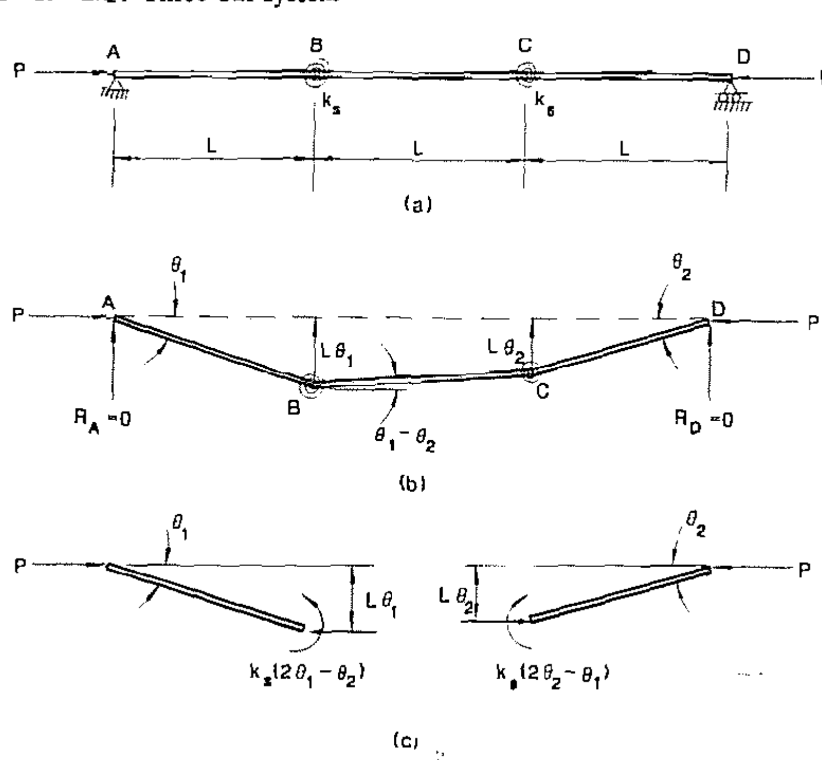
Figure 1.14a shows a three-bar system supported at the ends A and D by frictionless hinges and connected to one another at B and C by linear rotational springs of spring stiffness k_s. The system is assumed geometrically perfect in that the springs are unstrained when all the bars are in their horizontal orientation. We shall now determine the critical load P_{cr} of the system.

Bifurcation Approach

In using the bifurcation approach, we are investigating the equilibrium conditions of the system in a slightly deflected configuration. A configuration of the system displaced by an arbitrary kinematic admissible displacement is shown in Fig. 1.14b. A *kinematic admissible displacement* is a displacement that does not violate the constraints of the system. Note that the two parameters (θ_1 and θ_2) are necessary to describe fully the displaced configuration of the bars. Thus, the system is said to have two degrees of freedom.

It is clear from equilibrium consideration that the support reactions at

FIGURE 1.14 Three-bar system



A and D are zeros. By using the free body diagrams of the left and right bars, respectively (Fig. 1.14c), we can write the following equilibrium equations:

$$k_s(2\theta_1 - \theta_2) - P(L\theta_1) = 0, \tag{1.4.30a}$$

$$k_s(2\theta_2 - \theta_1) - P(L\theta_2) = 0. \tag{1.4.30b}$$

In matrix form, we have

$$\begin{bmatrix} 2k_s - PL & -k_s \\ -k_s & 2k_s - PL \end{bmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{1.4.31}$$

For a nontrivial solution, we must have the determinant of the coefficient matrix equal to zero

$$\det \begin{vmatrix} 2k_s - PL & -k_s \\ -k_s & 2k_s - PL \end{vmatrix} = (2k_s - PL)^2 - k_s^2 = 0 \tag{1.4.32}$$

Equation (1.4.32) is the characteristic equation of the system. The two eigenvalues are

$$P = \frac{k_s}{L} \tag{1.4.33a}$$

and

$$P = \frac{3k_s}{L}. \tag{1.4.33b}$$

The corresponding eigenvectors are

$$\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \tag{1.4.34a}$$

and

$$\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{1.4.34b}$$

The two deflected configurations of the system that correspond to the eigenvectors Eqs. (1.4.34a, b) are sketched in Fig. 1.15. Since the lowest value of the eigenvalue is the critical load of the system, we therefore have

$$P = P_{cr} = \frac{k_s}{L} \tag{1.4.35}$$

and the symmetric mode (Fig. 1.15a) is the buckling mode of the system.

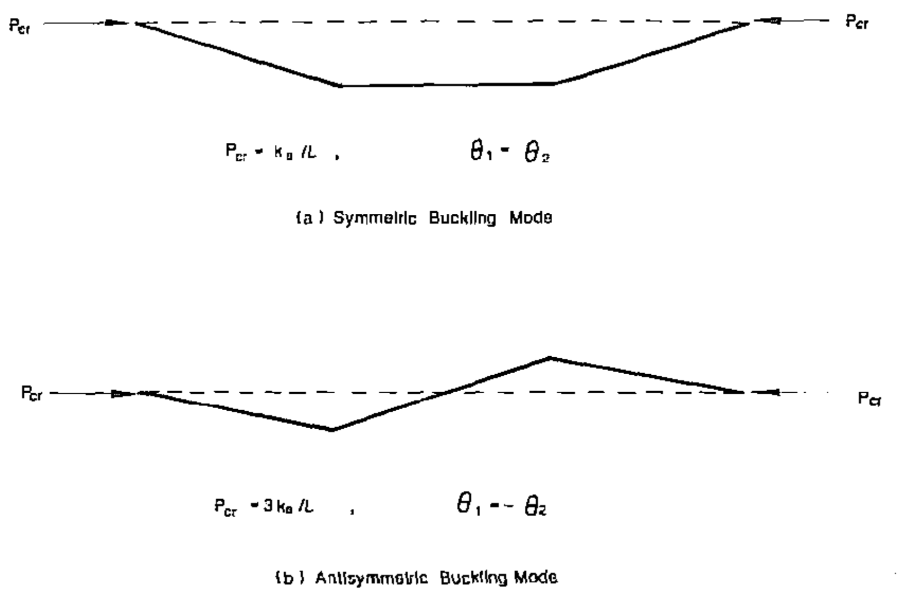


FIGURE 1.15 Buckling modes of the three-bar system

Energy Approach

For the system shown in Fig. 1.14, the strain energy is equal to the strain energy stored in the two springs.

$$U = \tfrac{1}{2}k_s(2\theta_1 - \theta_2)^2 + \tfrac{1}{2}k_s(2\theta_2 - \theta_1)^2 \tag{1.4.36}$$

The potential energy is equal to the negative of the work done by the external forces.

$$\begin{aligned} V &= -PL[(1 - \cos\theta_1) + (1 - \cos\theta_2) + 1 - \cos(\theta_1 - \theta_2)] \\ &= -PL[3 - \cos\theta_1 - \cos\theta_2 - \cos(\theta_1 - \theta_2)] \end{aligned} \tag{1.4.37}$$

The total potential energy of the system is equal to the sum of the strain energy and the potential energy.

$$\begin{aligned} \Pi &= U + V \\ &= \tfrac{1}{2}k_s(2\theta_1 - \theta_2)^2 + \tfrac{1}{2}k_s(2\theta_2 - \theta_1)^2 \\ &\quad - PL[3 - \cos\theta_1 - \cos\theta_2 - \cos(\theta_1 - \theta_2)] \end{aligned} \tag{1.4.38}$$

For equilibrium, the total potential energy of the system must be stationary. In mathematical terms, this requires

$$\begin{aligned} \frac{\partial\Pi}{\partial\theta_1} &= 2k_s(2\theta_1 - \theta_2) - k_s(2\theta_2 - \theta_1) \\ &\quad - PL[\sin\theta_1 + \sin(\theta_1 - \theta_2)] = 0 \end{aligned} \tag{1.4.39a}$$

$$\frac{\partial \Pi}{\partial \theta_2} = -k_s(2\theta_1 - \theta_2) + 2k_s(2\theta_2 - \theta_1)$$
$$- PL[\sin \theta_2 - \sin(\theta_1 - \theta_2)] = 0 \qquad (1.4.39b)$$

Upon simplification and using small angle approximation, we can write Eqs. (1.4.39a and b) in matrix form as

$$\begin{bmatrix} 5k_s - 2PL & -4k_s + PL \\ -4k_s + PL & 5k_s - 2PL \end{bmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad (1.4.40)$$

For nontrivial solution, we must have

$$\det \begin{vmatrix} 5k_s - 2PL & -4k_s + PL \\ -4k_s + PL & 5k_s - 2PL \end{vmatrix} = 0 \qquad (1.4.41)$$

from which the two eigenvalues are

$$P = \frac{k_s}{L} \qquad (1.4.42a)$$

or

$$P = \frac{3k_s}{L} \qquad (1.4.42b)$$

The smallest eigenvalue is the critical load of the system, therefore

$$P = P_{cr} = \frac{k_s}{L} \qquad (1.4.43)$$

To study the nature of equilibrium for the system in its undeflected ($\theta_1 = \theta_2 = 0$) position, we need to investigate higher order derivatives of the total potential energy function. By taking the second derivative of the total potential energy function, we have

$$\frac{\partial^2 \Pi}{\partial \theta_1^2} = 5k_s - PL[\cos \theta_1 + \cos(\theta_1 - \theta_2)]$$
$$\approx 5k_s - 2PL \qquad (1.4.44a)$$

$$\frac{\partial^2 \Pi}{\partial \theta_2^2} = 5k_s - PL[\cos \theta_2 + \cos(\theta_1 - \theta_2)]$$
$$\approx 5k_s - 2PL \qquad (1.4.44b)$$

$$\frac{\partial^2 \Pi}{\partial \theta_1 \, \partial \theta_2} = -4k_s + PL \cos(\theta_1 - \theta_2)$$
$$\approx -4k_s + PL \qquad (1.4.44c)$$

The equilibrium is stable if all of the following conditions are satisfied:

$$\frac{\partial^2\Pi}{\partial\theta_1^2} > 0 \tag{1.4.45a}$$

$$\frac{\partial^2\Pi}{\partial\theta_2^2} > 0 \tag{1.4.45b}$$

$$\left(\frac{\partial^2\Pi}{\partial\theta_1^2}\right)\left(\frac{\partial^2\Pi}{\partial\theta_2^2}\right) > \left(\frac{\partial^2\Pi}{\partial\theta_1\,\partial\theta_2}\right)^2 \tag{1.4.45c}$$

In view of Eqs. (1.4.44a–c), Equations (1.4.45a–c) become

$$P < \frac{5}{2}\left(\frac{k_s}{L}\right) \tag{1.4.46a}$$

$$P < \frac{5}{2}\left(\frac{k_s}{L}\right) \tag{1.4.46b}$$

$$(k_s - PL)(9k_s - 3PL) > 0 \tag{1.4.46c}$$

For $P < k_s/L$, all the inequalities expressed in Eqs. (1.4.46a–c) will be satisfied. Therefore for $P < k_s/L$, the equilibrium position $\theta_1 = \theta_2 = 0$ is stable, whereas for $P > k_s/L$ it is unstable.

1.5 ILLUSTRATIVE EXAMPLES—LARGE DEFLECTION ANALYSIS

In the foregoing analyses of the simple bar-spring models, the assumption of small deflection has been used because in these examples we are only interested in identifying the critical conditions and finding the critical loads of the system. Such an analysis is known as a *linear eigenvalue analysis*. Although, in addition to determining the critical loads, it is possible for us to investigate the nature of the equilibrium conditions of the systems in their *undeflected* configurations by studying the second derivatives of the total potential energy functions, a linear eigenvalue analysis can provide us with no information about the behavior of the systems after the critical loads have been reached. In other words, if the analysis is performed using the small displacement assumption, it is not possible to study the *postbuckling behavior* of the system. To study the postbuckling behavior of a system, we must use large displacement analysis. This is illustrated in the following examples.

1.5.1 Rigid Bar Supported by a Rotational Spring

Consider the simple one degree of freedom spring-bar system shown in Fig. 1.16. This simple model has been analyzed earlier using the small displacement assumption. The critical load was found to be $P_{cr} = k_s L$ and

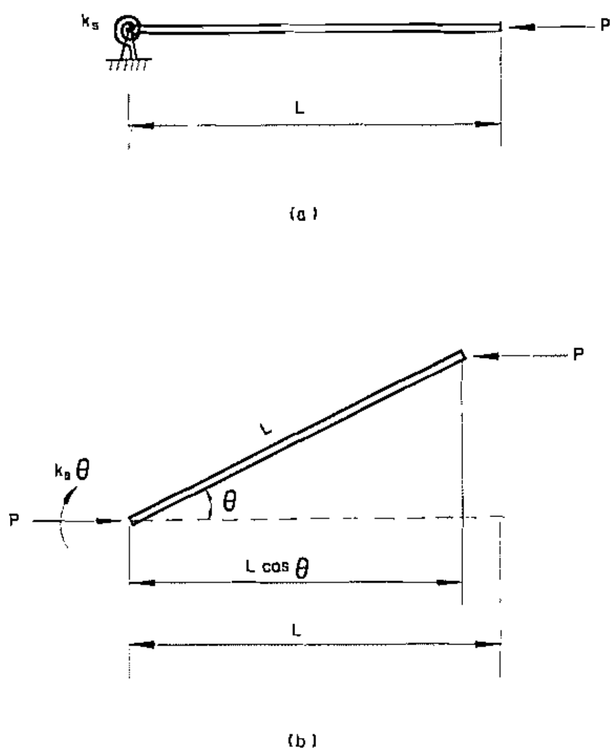


FIGURE 1.16 Rotational spring-supported rigid bar system (large deflection analysis)

by studying the nature of the second derivative of the total potential energy function, we concluded that the equilibrium position that corresponds to the initial (straight) configuration of the bar was stable if $P < P_{cr}$, but it became unstable if $P > P_{cr}$ (Fig. 1.11). However, no information about the nature of equilibrium can be obtained when $P = P_{cr}$ nor do we have any knowledge about the postbuckling behavior of the system. To obtain such information, it is necessary to perform a *large displacement analysis* as shown in the following.

Energy Approach

Although we could readily use the bifurcation approach to determine both the equilibrium paths of a system in a large displacement analysis and the critical load obtained at the point of intersection of the fundamental and postbuckling path(s), we will use the energy approach instead because, in addition to obtaining the critical load, with this second approach we can also investigate the stability of the postbuckling equilibrium paths of the system by examining the characteristic of the higher order derivatives of the total potential energy function.

The strain energy of the system is equal to the strain energy of the spring

$$U = \tfrac{1}{2} k_s \theta^2 \tag{1.5.1}$$

The potential energy of the system is the potential energy of the external force

$$V = -PL(1 - \cos\theta) \tag{1.5.2}$$

The total potential energy of the system is then

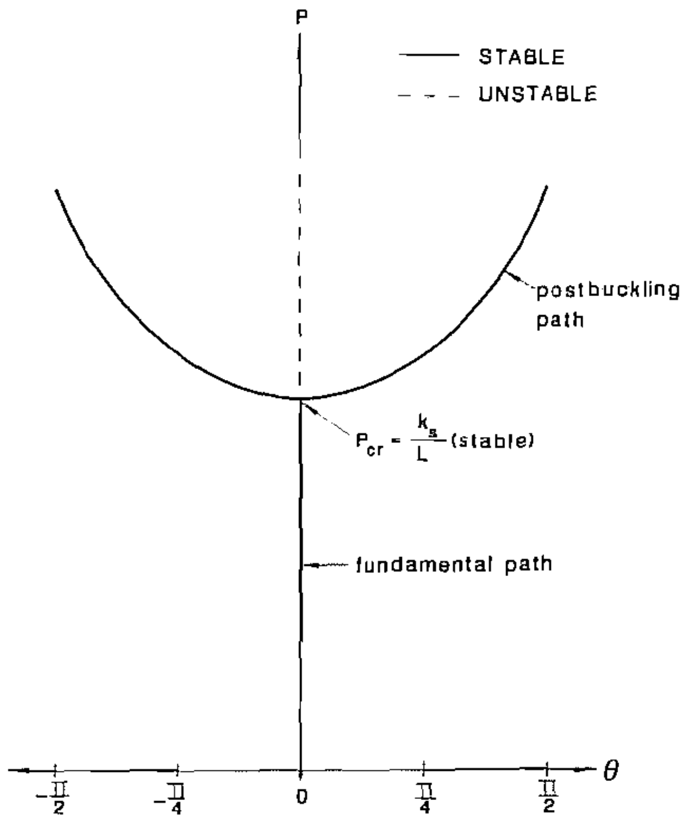
$$\Pi = U + V = \tfrac{1}{2} k_s \theta^2 - PL(1 - \cos\theta) \tag{1.5.3}$$

For equilibrium, we must have

$$\frac{d\Pi}{d\theta} = k_s \theta - PL \sin\theta = 0 \tag{1.5.4}$$

This equilibrium equation is satisfied for all values of P if $\theta = 0$. This trivial equilibrium path is the fundamental equilibrium path. It is plotted in Fig. 1.17. Note that this path is coincident with the load axis. The

FIGURE 1.17 Equilibrium paths of the spring-bar system shown in Fig. 1.16



postbuckling path is given by

$$P = \frac{k_s \theta}{L \sin \theta} \tag{1.5.5}$$

This path is also plotted in Fig. 1.17 for the range $-\pi/2 \leq \theta \leq \pi/2$. It intersects the fundamental path at $P_{cr} = k_s/L$.

To study the stability of the equilibrium paths, we need to examine the higher order derivatives of the total potential energy function. By taking the second derivative of Π, we have

$$\frac{d^2\Pi}{d\theta^2} = k_s - PL \cos \theta \tag{1.5.6}$$

For the fundamental path, $\theta = 0$, therefore Eq. (1.5.6) becomes

$$\frac{d^2\Pi}{d\theta^2} = k_s - PL \tag{1.5.7}$$

The quantity $d^2\Pi/d\theta^2$ changes from positive to negative at $P = P_{cr} = k_s/L$, indicating that the initial horizontal position of the bar is stable if $P < P_{cr}$ but unstable if $P > P_{cr}$.

For the postbuckling path, $P = k_s\theta/L \sin \theta$, therefore Eq. (1.5.6) becomes

$$\begin{aligned} \frac{d^2\Pi}{d\theta^2} &= k_s - \left(\frac{k_s \theta}{L \sin \theta}\right) L \cos \theta \\ &= k_s\left(1 - \frac{\theta}{\tan \theta}\right) \end{aligned} \tag{1.5.8}$$

The quantity $d^2\Pi/d\theta^2$ is always positive since the quantity in parenthesis is always greater than zero, indicating that the postbuckling path is always stable.

At the critical point ($P = P_{cr} = k_s/L$), $d^2\Pi/d\theta^2$ is zero according to Eq. (1.5.7), so no information concerning the stability of the system can be obtained. To investigate the stability of this critical equilibrium state, we need to examine the first nonzero term in a Taylor series expansion for Π. Using a Taylor series expansion for the total potential energy function about $\theta = 0$, we obtain

$$\begin{aligned} \Pi = \Pi\Big|_{\theta=0} &+ \frac{d\Pi}{d\theta}\Big|_{\theta=0} \theta + \frac{1}{2}\frac{d^2\Pi}{d\theta^2}\Big|_{\theta=0} \theta^2 \\ &+ \frac{1}{6}\frac{d^3\Pi}{d\theta^3}\Big|_{\theta=0} \theta^3 + \frac{1}{24}\frac{d^4\Pi}{d\theta^4}\Big|_{\theta=0} \theta^4 + \cdots \end{aligned} \tag{1.5.9}$$

It can easily be shown that the first four terms in Eq. (1.5.9) are zeros, thus the first nonzero term of the series is the fifth term

$$\frac{1}{24}\frac{d^4\Pi}{d\theta^4}\bigg|_{\theta=0}\theta^4=\frac{1}{24}PL\theta^4 \tag{1.5.10}$$

At $P = P_{cr} = k_s/L$, we have

$$\frac{1}{24}PL\theta^4=\frac{1}{24}k_s\theta^4 \tag{1.5.11}$$

which is positive, indicating that Π is a local minimum and so the equilibrium state at the critical point is stable.

From the above discussion it can be seen that as P increases gradually to P_{cr}, the stable fundamental equilibrium path bifurcates into an unstable equilibrium path corresponding to the original horizontal position of the bar and a stable postbuckling equilibrium path corresponding to the deflected configuration of the bar. The equilibrium state at the critical point is stable. The stable postbuckling equilibrium path is symmetric about the load axis, indicating that the bar can deflect either upward or downward with no particular preference.

1.5.2 Rigid Bar Supported by a Translational Spring

Consider now the one degree of freedom bar-spring system shown in Fig. 1.18. This model has been analyzed earlier using the small displacement assumption. The critical load was found to be $P_{cr}=k_sL$. To study the postbuckling behavior, we must use the large deflection analysis.

Energy Approach

Here, as in the previous example, the energy approach is used. The strain energy of the system is

$$U=\tfrac{1}{2}k_s(L\sin\theta)^2 \tag{1.5.12}$$

The potential energy of the system is

$$V=-PL(1-\cos\theta) \tag{1.5.13}$$

The total potential energy of the system is

$$\Pi=U+V=\tfrac{1}{2}k_s(L\sin\theta)^2-PL(1-\cos\theta) \tag{1.5.14}$$

For equilibrium, we must have

$$\begin{aligned}\frac{d\Pi}{d\theta}&=k_sL^2\sin\theta\cos\theta-PL\sin\theta\\&=(k_sL^2\cos\theta-PL)\sin\theta=0\end{aligned} \tag{1.5.15}$$

This equilibrium equation is satisfied for all values of P when $\theta=0$,

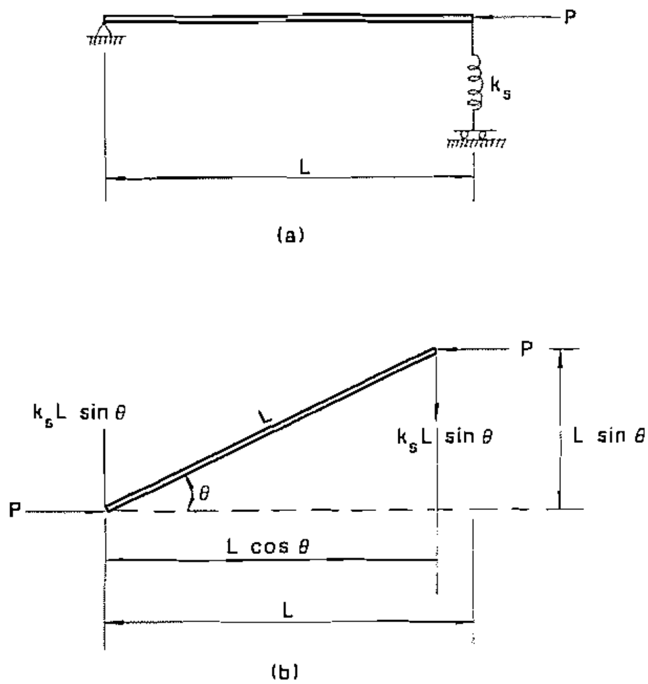


FIGURE 1.18 Translational spring-supported rigid bar system (large deflection analysis)

which is the fundamental equilibrium path (Fig. 1.19). The postbuckling path is given by

$$P = k_s L \cos\theta \tag{1.5.16}$$

The fundamental and postbuckling paths are plotted in Fig. 1.19 for the range $-\pi/2 \leqslant \theta \leqslant \pi/2$. They intersect at $P = P_{cr} = k_s L$.

To study the stability of the equilibrium paths, we form

$$\frac{d^2\Pi}{d\theta^2} = k_s L^2(\cos^2\theta - \sin^2\theta) - PL\cos\theta \tag{1.5.17}$$

For the fundamental path, $\theta = 0$, Eq. (1.5.17) thus becomes

$$\frac{d^2\Pi}{d\theta^2} = k_s L^2 - PL \tag{1.5.18}$$

The quantity $d^2\Pi/d\theta^2$ changes from positive to negative at $P = P_{cr} = k_s L$, which indicates that the initial horizontal position of the bar is stable if $P < P_{cr}$ but unstable if $P > P_{cr}$.

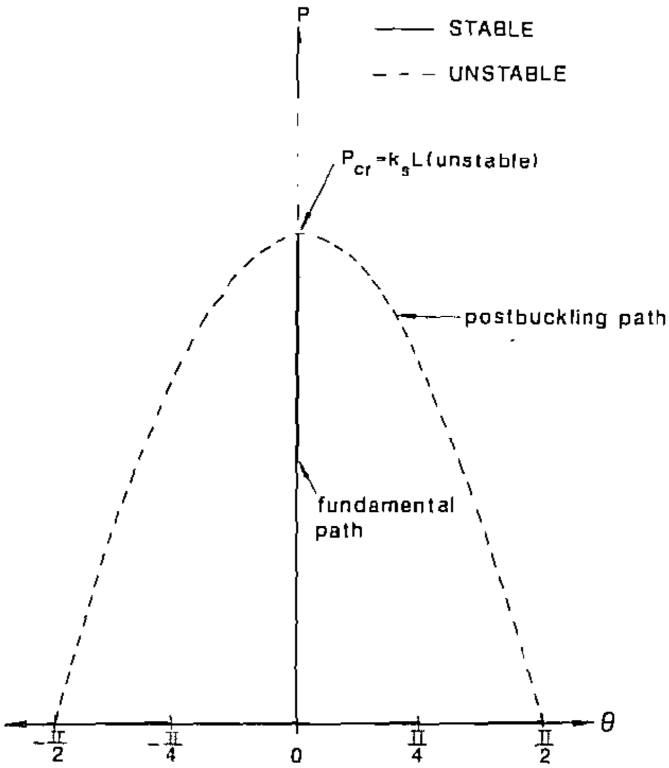


FIGURE 1.19 Equilibrium paths of the spring-bar system shown in Fig. 1.18

For the postbuckling path, $P = k_sL \cos\theta$, Eq. (1.5.17) thus becomes

$$\frac{d^2\Pi}{d\theta^2} = k_sL^2(\cos^2\theta - \sin^2\theta) - (k_sL\cos\theta)L\cos\theta$$
$$= -k_sL^2\sin^2\theta \tag{1.5.19}$$

The quantity $d^2\Pi/d\theta^2$ is always negative, indicating that the postbuckling path is unstable.

At the critical point ($P = P_{cr} = k_sL$), $d^2\Pi/d\theta^2$ is zero according to Eq. (1.5.18). As a result, we need to expand the total potential energy function in a Taylor series and examine the first nonzero term in the series. If we substitute the expression for Π and its derivatives into the Taylor series, Eq. (1.5.9), it can be shown that the first nonzero term is

$$\frac{1}{24}\frac{d^4\Pi}{d\theta^4}\bigg|_{\theta=0}\theta^4 = \frac{1}{24}(-4k_sL^2 + PL)\theta^4 \tag{1.5.20}$$

At $P = P_{cr} = k_s L$, we have

$$\frac{1}{24}(-4k_sL^2 + PL)\theta^4 = -\tfrac{1}{8}k_sL^2\theta^4 \qquad (1.5.21)$$

which is negative, indicating that Π is a local maximum and so the equilibrium state at the critical point is unstable.

From the above discussion, it can be seen that as P increases gradually to P_{cr}, the stable fundamental equilibrium path bifurcates into an unstable equilibrium path corresponding to the original horizontal position of the bar and an unstable equilibrium path corresponding to the deflected configuration of the bar. The equilibrium state at the critical point is also unstable.

1.6 ILLUSTRATIVE EXAMPLES—IMPERFECT SYSTEMS

Note that for all the examples presented in the preceding sections it has been assumed that the systems are geometrically *perfect*. When the bars are in their horizontal positions, the springs are unstrained at the commencement of the loadings. The systems will therefore remain undeflected until the values of P reach their critical values, P_{cr}. Suppose now that the systems are *imperfect* in the sense that the bars are slightly tilted when the springs are unstrained. The bar will deflect as soon as the load is applied. The problem then becomes a *load-deflection problem*. The following examples will illustrate the effect of this imperfection on the response of the systems.

1.6.1 Rigid Bar Supported by a Rotational Spring

Consider the one-degree-of-freedom-imperfect-bar-spring system shown in Fig. 1.20. The system is imperfect in that the bar is tilted slightly by an angle θ_0, and at this tilted position the spring is unstretched. We shall now study the behavior of the system using the energy approach.

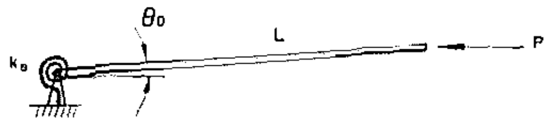
Energy Approach

The strain energy of the system is equal to the strain energy stored in the spring

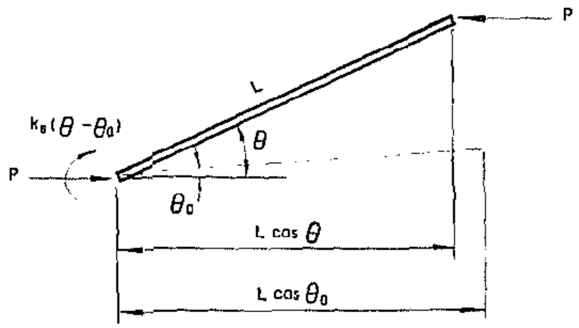
$$U = \tfrac{1}{2}k_s(\theta - \theta_0)^2 \qquad (1.6.1)$$

The potential energy of the system is equal to the potential energy of the external force

$$V = -PL(\cos\theta_0 - \cos\theta) \qquad (1.6.2)$$



(a)



(b)

FIGURE 1.20 Rotational spring-supported imperfect rigid bar system

The total potential energy of the system is

$$\Pi = U + V = \tfrac{1}{2}k_s(\theta - \theta_0)^2 - PL(\cos\theta_0 - \cos\theta) \tag{1.6.3}$$

For equilibrium, the total potential energy of the system must be stationary

$$\frac{d\Pi}{d\theta} = k_s(\theta - \theta_0) - PL\sin\theta = 0 \tag{1.6.4}$$

from which, we obtain

$$P = \frac{k_s(\theta - \theta_0)}{L\sin\theta} \tag{1.6.5}$$

The equilibrium paths given in Eq. (1.6.5) for $\theta_0 = 0.1$ and 0.3 are plotted in Fig. 1.21. The figure also shows the equilibrium paths for the corresponding perfect system that was discussed earlier (Fig. 1.17). For the *imperfect* system, deflection commences as soon as the load is

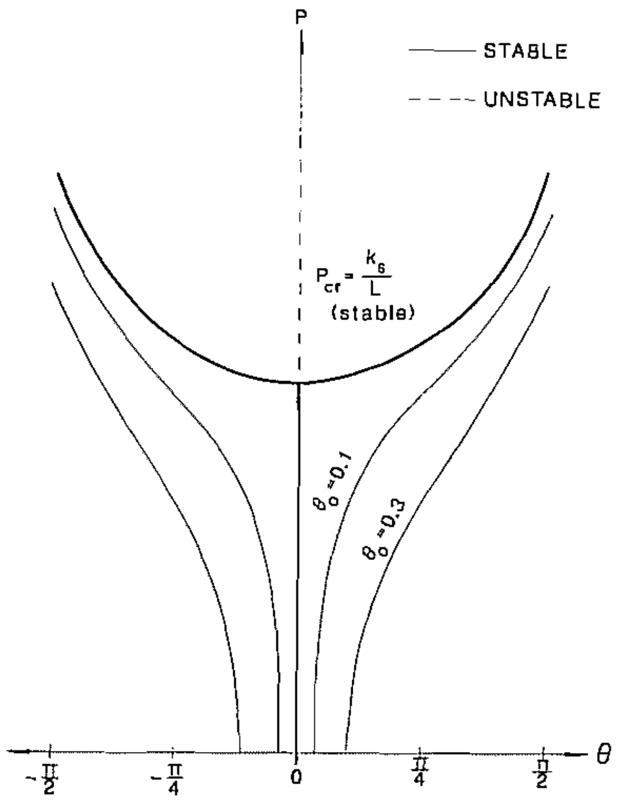


FIGURE 1.21 Equilibrium paths of the imperfect spring-bar system shown in Fig. 1.20

applied. The smaller the imperfection, the closer the equilibrium paths of the imperfect system is to that of the perfect system. In fact, if the imperfection vanishes, the equilibrium paths of the imperfect system will collapse onto the equilibrium paths of the perfect system.

To study the stability of the equilibrium paths of the imperfect system, we need to examine the second derivative of the total potential energy function

$$\frac{d^2\Pi}{d\theta^2} = k_s - PL\cos\theta \tag{1.6.6}$$

Therefore, the equilibrium paths are stable if

$$P < \frac{k_s}{L\cos\theta} \tag{1.6.7}$$

and they are unstable if

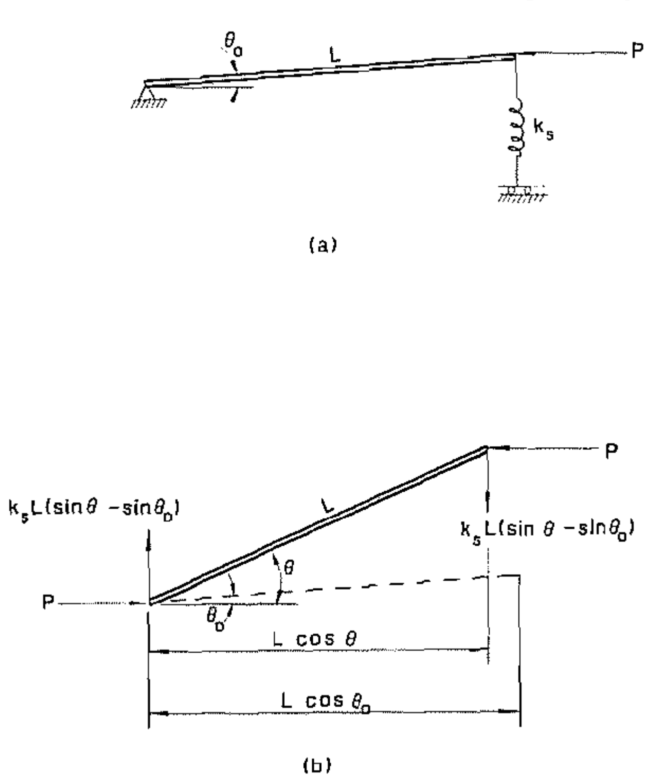
$$P > \frac{k_s}{L\cos\theta} \tag{1.6.8}$$

It is quite obvious that Eq. (1.6.7) always controls for $-\pi/2 \leq \theta \leq \pi/2$. Therefore, the equilibrium paths described by Eq. (1.6.5) are always stable, indicating that as P increases the deflections increase, as shown by the curves with the initial angle $\theta_0 = 0$, 0.1 and 0.3, and no instability will occur. The maximum load that the system can carry is greater than the critical load, P_{cr}.

1.6.2 Rigid Bar Supported by a Translation Spring

Consider now the imperfect system shown in Fig. 1.22. The system is imperfect, being tilted by an angle θ_0 when the spring is unstretched. We shall now use the energy approach to study the response of this imperfect system.

FIGURE 1.22 Translational spring-supported imperfect rigid bar system



Energy Approach

The strain energy of the system is equal to the strain energy stored in the spring

$$U = \tfrac{1}{2}k_s L^2(\sin\theta - \sin\theta_0)^2 \tag{1.6.9}$$

The potential energy of the system is equal to the potential energy of the external force

$$V = -PL(\cos\theta_0 - \cos\theta) \tag{1.6.10}$$

The total potential energy of the system is

$$\Pi = U + V = \tfrac{1}{2}k_s L^2(\sin\theta - \sin\theta_0)^2 - PL(\cos\theta_0 - \cos\theta) \tag{1.6.11}$$

For equilibrium, we must have

$$\frac{d\Pi}{d\theta} = k_s L^2(\sin\theta - \sin\theta_0)\cos\theta - PL\sin\theta = 0 \tag{1.6.12}$$

from which, we obtain

$$P = k_s L\cos\theta\left(1 - \frac{\sin\theta_0}{\sin\theta}\right) \tag{1.6.13}$$

The equilibrium paths described by Eq. (1.6.13) for $\theta_0 = 0.1$ and 0.3 are plotted in Fig. 1.23. The equilibrium paths for the corresponding perfect system are also shown in the figure. Again, as in the preceding example, as the imperfection θ_0 approaches zero, the equilibrium paths of the imperfect system collapse onto the equilibrium paths of the perfect system. The maximum loads P_{max} (the peak points of the load-deflection curves) that the imperfect system can carry are less than the critical load P_{cr} of the corresponding perfect system. These maximum loads can be evaluated by setting

$$\frac{dP}{d\theta} = k_s L\left(-\sin\theta + \frac{\sin\theta_0}{\sin^2\theta}\right) = 0 \tag{1.6.14}$$

from which we obtain the condition

$$\sin\theta_0 = \sin^3\theta \tag{1.6.15}$$

Substitution of Eq. (1.6.15) into Eq. (1.6.13) gives

$$P = P_{max} = k_s L\cos^3\theta \tag{1.6.16}$$

The locus of the maximum loads as described by Eq. (1.6.16) is plotted in Fig. 1.23 as a dash-and-dotted line. Note that P_{max} is always less than P_{cr}, except at $\theta = 0$, when P_{max} becomes P_{cr}. The larger the imperfection the smaller P_{max} will be.

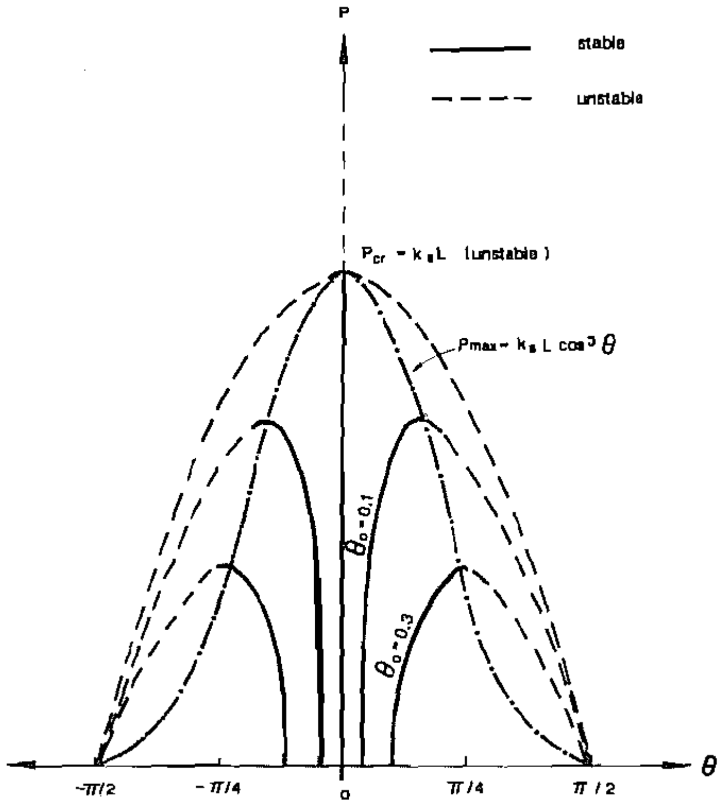


FIGURE 1.23 Equilibrium paths of the imperfect spring-bar system shown in Fig. 1.22

To study the stability of the equilibrium paths described by Eq. (1.6.13), we need to examine the second derivative of the total potential energy function.

$$\frac{d^2\Pi}{d\theta^2} = k_s L^2(\cos^2\theta - \sin^2\theta + \sin\theta_0 \sin\theta) \\ - PL\cos\theta \tag{1.6.17}$$

Upon substitution of Eq. (1.6.13) for P into the above equation and simplifying, we obtain

$$\frac{d^2\Pi}{d\theta^2} = k_s L^2\left(\frac{\sin\theta_0 - \sin^3\theta}{\sin\theta}\right) \tag{1.6.18}$$

Thus, for the range $-\pi/2 \leq \theta \leq \pi/2$, the equilibrium paths given by Eq. (1.6.13) will be stable if

$$\sin\theta_0 > \sin^3\theta \tag{1.6.19}$$

and they will be unstable if

$$\sin \theta_0 < \sin^3 \theta \qquad (1.6.20)$$

So, it can be seen that the rising equilibrium paths are stable and the falling equilibrium paths are unstable. As the load P is increased from zero, the load-deflection behavior of the imperfect system will follow the stable rising equilibrium paths as shown in Fig. 1.23, until P_{max} is reached, after which the system will become unstable.

From the foregoing examples of imperfect systems, we can conclude that for a system with stable postbuckling equilibrium paths, a small imperfection will not significantly affect the system's behavior. The maximum load the imperfect system can carry is larger than the critical load of the perfect system. On the other hand, for a system with unstable postbuckling equilibrium paths, a small imperfection may have a noticeable effect on the system's behavior. The maximum load that the imperfect system can carry is then smaller than the critical load of the perfect system, and the magnitude of this maximum load decreases with increasing imperfection.

1.7 DESIGN PHILOSOPHIES

To implement the mathematical theory of stability into engineering practice, it is necessary to review the various design philosophies and safety concepts upon which current design practice is based. Details of this implementation on various specific subjects will be given in the chapters that follow.

At present, design practice is based on one of these three design philosophies: Allowable Stress Design, Plastic Design, and Load and Resistance Factor Design. A brief discussion of these design philosophies will be given in the following sections.

1.7.1 Allowable Stress Design

The purpose of allowable stress design (ASD) is to ensure that the stresses computed under the action of the working, i.e., service loads, of a structure do not exceed some predesignated *allowable* values. The allowable stresses are usually expressed as a function of the yield stress or ultimate stress of the material. The general format for an allowable stress design is thus

$$\frac{R_n}{F.S.} \geq \sum_{i=1}^{m} Q_{ni} \qquad (1.7.1)$$

where

R_n = nominal resistance of the structural member expressed in unit of stress
Q_n = nominal working, or service stresses computed under working load conditions
$F.S.$ = factor of safety
i = type of load (i.e., dead load, live load, wind load, etc.)
m = number of load type

The left hand side of Eq. (1.7.1) represents the allowable stress of the structural member or component under various loading conditions (for example, tension, compression, bending, shear, etc.). The right hand side of the equation represents the combined stress produced by various load combinations (for example, dead load, live load, wind load, etc.). Formulas for the allowable stresses for various types of structural members under various types of loadings are specified in the AISC Specification.[3] A satisfactory design is when the stresses in the member computed using a first-order analysis under working load conditions do not exceed their allowable values. One should realize that in allowable stress design, the factor of safety is applied to the resistance term, and safety is evaluated in the service load range.

1.7.2 Plastic Design

The purpose of plastic design (PD) is to ensure that the maximum plastic strength of the structural member or component does not exceed that of the factored load combinations. It has the format

$$R_n \geqslant \gamma \sum_{i=1}^{m} Q_{ni} \tag{1.7.2}$$

where

R_n = nominal plastic strength of the member
Q_n = nominal load effect (e.g., axial force, shear force, bending moment, etc.)
γ = load factor
i = type of load
m = number of load types

In steel design, the load factor is designated by the AISC Specification[3] as 1.7 if Q_n consists of only dead and live gravity loads, or as 1.3 if Q_n consists of dead and live gravity loads plus wind or earthquake loads. The use of a smaller load factor for the latter case is justified by the fact that the simultaneous occurrence of all these load effects is not very likely.

Note that in plastic design, safety is incorporated in the load term and is evaluated at the ultimate (plastic strength) limit state.

1.7.3 Load and Resistance Factor Design

The purpose of load and resistance factor design (LRFD) is to ensure that the nominal resistance of the structural member or component exceeds that of the load effects. Two safety factors are used, one applied to the loads, and the other to the resistance of the materials. Thus, the load and resistance factor design has the format

$$\phi R_n \geq \sum_{i=1}^{m} \gamma_i Q_{ni} \tag{1.7.3}$$

where

R_n = nominal resistance of the structural member
Q_n = load effect (e.g., axial force, shear force, bending moment, etc.)
ϕ = resistance factor (≤ 1.0)
γ_i = load factor (usually >1.0) corresponding to Q_{ni}
i = type of load
m = number of load types

In the 1986 LRFD Specification,[1] the resistance factors were developed mainly through calibration,[4] whereas the load factors were developed based on statistical analysis.[5,6] In particular, the first-order probability theory is used. The load and resistance factors for various types of loadings and various load combinations are summarized in Tables 1.1 and 1.2, respectively. A satisfactory design is the one in which the probability of the structural member exceeding a limit state (for example, yielding, fracture, buckling, etc.) is minimal. Based on the first-order second-moment probabilistic analysis,[7] the safety of the structural member is measured by a *reliability* or *safety index*[4] defined as

$$\beta = \frac{\ln(\bar{R}_n/\bar{Q}_n)}{\sqrt{V_R^2 + V_Q^2}} \tag{1.7.4}$$

where

$\bar{R}$ = mean resistance
$\bar{Q}$ = mean load effect
V_R = coefficient of variation of resistance
$= \dfrac{\sigma_R}{\bar{R}}$
V_Q = coefficient of variation of load effect
$= \dfrac{\sigma_Q}{\bar{Q}}$

in which σ equals the standard deviation.

Table 1.1 Load Factors and Load Combinations[a]

$1.4\ D$

$1.2\ D + 1.6\,L + 0.5\,(L_r \text{ or } S \text{ or } R)$

$1.2\ D + 1.6\,(L_r \text{ or } S \text{ or } R) + (0.5\,L \text{ or } 0.8\,W)$

$1.2\ D + 1.3\,W + 0.5\,L + 0.5\,(L_r \text{ or } S \text{ or } R)$

$1.2\ D + 1.5\,E + (0.5\,L \text{ or } 0.2\,S)$

$0.9\ D - 1.3\,W \text{ or } 1.5\,E$

where

D = dead load
L = live load
L_r = roof live load
W = wind load
S = snow load
E = earthquake load
R = nominal load due to initial rainwater or ice exclusive of the ponding contribution

[a] The load factor on L in the third, fourth and fifth load combinations shown above shall equal 1.0 for garages, areas occupied as places of public assembly and all areas where the live load is greater than 100 psf.

The physical interpretation of the reliability index β is shown in Fig. 1.24. It is the multiplier of the standard deviation $\sqrt{V_R^2 + V_Q^2}$ between the mean of the $\ln(R/Q)$ distribution and the ordinate. Note that both the resistance R and the load Q are treated as random parameters in LRFD, and so $\ln(R/Q)$ does not have a single value but follows a distribution. The shaded area in the figure represents the probability in which $\ln(R/Q) < 1$, i.e., the probability that the resistance will be smaller than

Table 1.2 Resistance Factors

Member type and limit state	ϕ
Tension member, limit state: yielding	0.90
Tension member, limit state: fracture	0.75
Pin-connected member, limit state: tension	0.75
Pin-connected member, limit state: shear	0.75
Pin-connected member, limit state: bearing	1.00
Columns, all limit states	0.85
Beams, all limit states	0.90
High-strength bolts, limit state: tension	0.75
High-strength bolts, limit state: shear	
A307 bolts	0.60
Others	0.65

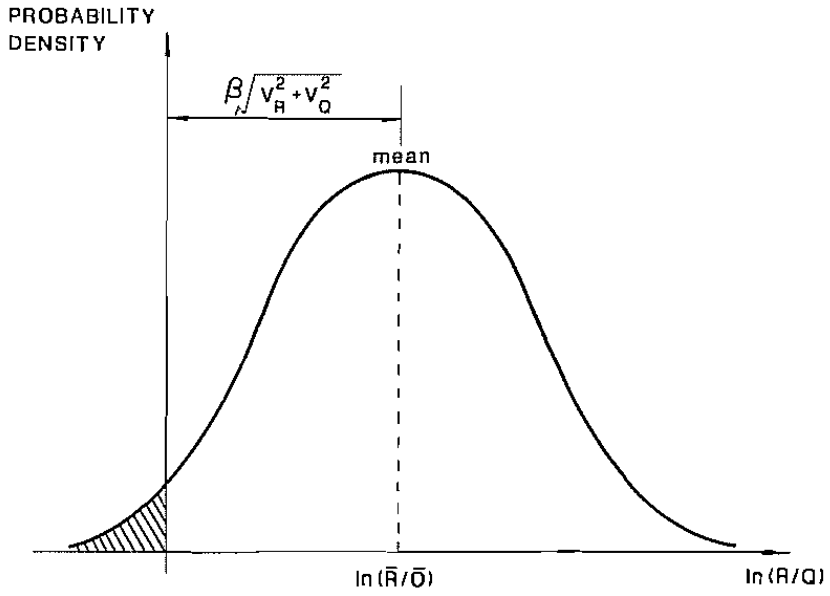


FIGURE 1.24 Reliability index

the load effect, indicating that a limit state has been exceeded. The larger the value of β, the smaller the area of the shaded area, and so it becomes more improbable that a limit state may be exceeded. Thus, the magnitude of β reflects the safety of the member.

In the development of the present LRFD Specification,[1] the following target values for β were selected:

1. $\beta = 3.0$ for members and $\beta = 4.5$ for connectors under dead plus live and/or snow loading;
2. $\beta = 2.5$ for members under dead plus live load acting in conjunction with wind loading, and;
3. $\beta = 1.75$ for members under dead plus live load acting in conjunction with earthquake loading.

A higher value of β for connectors ensures that the connections designed are stronger than their adjoining members. A lower value of β for members under the action of a combination of dead, live, wind, or earthquake loading reflects the improbability that these loadings will act simultaneously.

From the above discussion, it can be seen that in allowable stress design, the safety of the structural member is evaluated on the basis of service load conditions, whereas in plastic or load and resistance factor design, safety is evaluated on the basis of the ultimate or limit load conditions. In addition to strength, the designer must also pay attention to stiffness of the structure. One important criterion related to stiffness is that the structure or structural component must not deflect excessively under service load conditions. Thus, regardless of the design method, one should always investigate such *serviceability requirements* as deflection and vibration at service load conditions.

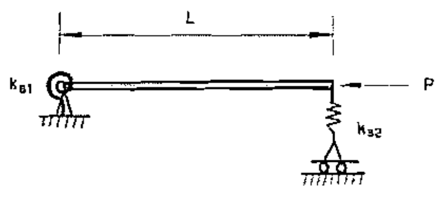
1.8 SCOPE

In this book, the discussion of *stability theory* will be limited to conservative systems under static or quasistatic loads. A *conservative system* is a system that is subjected only to conservative forces, that is, to forces whose potential energy is dependent only on the final values of deflection. In addition, most of the material presented in this book's later chapters will be based on what is called *small displacement theory*. Hence, the critical load but not the postbuckling behavior of the member or structure will be studied. Only the stability behavior of structural members and frames will be presented; the stability of plates and shells will not be discussed.

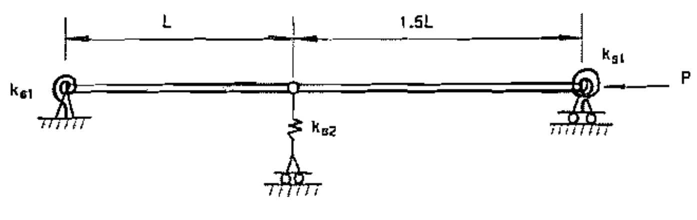
For a discussion of the stability behavior of elastic systems under nonconservative and dynamic forces, interested readers should refer to books by Bolotin.[8,9] For a discussion of the buckling behavior of plates and shells, please refer to books by Timoshenko and Gere[10] and Brush and Almroth.[11]

PROBLEMS

1.1 Find the critical load P_{cr} of the bar-spring systems shown in Fig. P1.1a–c using the bifurcation approach. Assume that all the bars are rigid.



(a)



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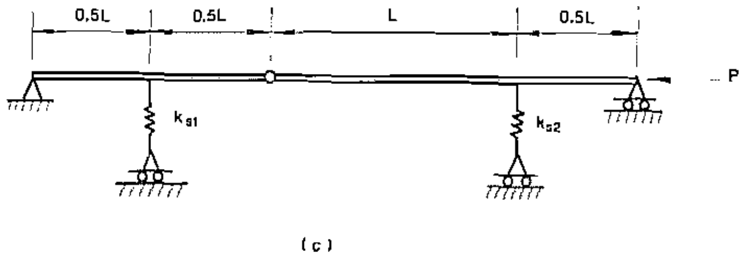

FIGURE P1.1

1.2 Repeat Prob. 1.1, using the energy approach.

1.3 Investigate the stability behavior of the asymmetric spring-bar model shown in Fig. P1.3.

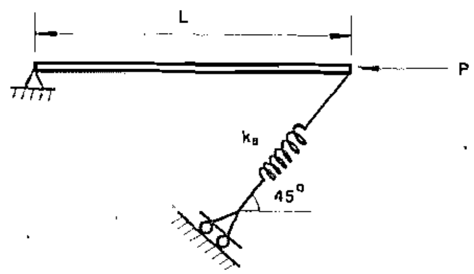

FIGURE P1.3

1.4 Investigate the stability behavior of the snap-through spring-bar model shown in Fig. P.1.4.

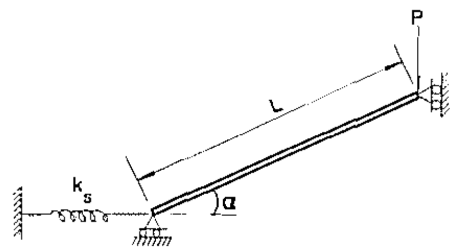

FIGURE P1.4

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